SE(3)-WAM: What Should a Robot Remember and Predict for Dynamic Manipulation? Supplementary Material

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This supplement gives the data, model interfaces, experimental protocols, and full results. A–D cover the benchmark, history, and future representations; E defines backbone and information-flow comparisons; F describes robot-data pretraining and deployment; G–H analyze predictions, failures, and execution. An em dash denotes an unavailable measurement.

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A. Tasks, Data, and Evaluation

A.1. The Fourteen-Task Suite

The design studies use the DB0 v0.6 demonstration pool and its fourteen-task evaluation suite. Table S1 lists the task identifiers used in the result arrays. The task-balanced score includes every task, including zero-success conditions.

The common training pool has 11,200 episodes, 800 per task. The internal representation screen uses 350 validation episodes; adapted systems retain their own validation settings (including 1,400 for DreamVLA/OpenWAM). The future screen samples 160,416 chunks per epoch. Stored episode count and effective chunk exposure measure different aspects of training. Motion-only P0 prediction has no corresponding action-learning score in the fourteen-task table.

Success is read from the native task evaluator after its ordered physical requirements (Table S1). The accepted FUTURE0, BASE2, and HISTORY1 packages reference the same archived evaluator source, task configurations, and fourteen-suite scenario manifest. We recover the definitions from that archive. Success is checked at each 2 ms physics step. Interrupted continuous conditions reset their counters. For T2, the sample completing the low-relative-motion counter also starts the stable-dwell counter; the two durations should not be added as an exact 0.600 s requirement.

Terminal events. The native result mapping groups events into five categories: *invalid* (invalid state); *boundary* (workspace or downstream exit, drop, expired goal window); *safety* (unsafe contact, excessive rim impulse, wire contact, stage block, release impulse, early robot contact, tray or gripper–support contact, striker trap); *task violation* (blocked driver, release outside the goal, path deviation or backtracking, premature release, stopping/deflecting without capture, unstable tipping, missing/extra impact, wrong striker contact, or insufficient outgoing motion change); and *timeout*. These categories describe evaluator outcomes.

There is no global priority among categories. Each family selects the first terminal event in its physics-step evaluation order and preserves that result. Grasp checks invalid state, workspace exit, downstream exit, and drop before success, then timeout. T1 inserts unsafe occluder contact after invalid state. T2 checks invalid state, rim impulse $> 0.10\,\mathrm{N\,s}$, drop, goal-window expiry, and release outside the goal before success and timeout. T3 checks invalid state, stage block, drop, workspace exit, wire contact, premature release, path deviation, and backtracking before success and timeout. Wire failure uses impulse $> 0.020\,\mathrm{N\,s}$ or penetration $> 0.004\,\mathrm{m}$ for more than two consecutive steps; path deviation is $> 0.030\,\mathrm{m}$ for 0.050 s, and backtracking exceeds 0.05 relative to maximum progress.

T4 checks invalid state, release impulse $> 0.002\,\mathrm{N\,s}$, unsafe ramp contact, unsupported tray contact, can tipping, early-contact/boundary branches, and invalid supported capture before success and timeout. A target that has exited the ramp, remains in contact with the downstream support, and has no fingerpad contact at speed $\leq 0.03\,\mathrm{m/s}$ for 0.25 s

Table S1. Native success conditions in the reported fourteen-task suite. Thresholds come from the evaluator source and task configurations bound to the accepted FUTURE0, BASE2, and HISTORY1 packages. A task must pass its preceding stages and avoid terminal failures. All tasks run at 20 Hz with a 2 ms physics step; durations below refer to physical time. COM denotes center of mass.

Task	Ordered success conditions	Limit
grasp; t1_belt, t1_xyz, t1_so3, t1_occ	Approach within 0.10 m; bilateral fingerpad contact for 0.05 s; collision-geometry bottom at least 0.05 m above support; retain contact and clearance for 0.50 s. T1 adds conveyor drive, translation, rotation, or a physical occluder. SO3 has no additional success-angle threshold; unsafe occluder contact terminates OCC.	160 steps 8 s
t2_trans	Approach within 0.10 m. Entire object collision geometry inside the receptacle: planar clearance and floor gap both ≥ -0.002 m, top no higher than the opening. Release requires an opening command < -0.5 and no fingerpad contact. Relative linear/angular speeds ≤ 0.08 m/s and 0.8 rad/s for 0.10 s, followed by a 0.50 s stable dwell.	200 10 s
t2_se3	Approach within 0.05 m and 0.12 rad. Full T-shaped containment: planar clearance ≥ 0.002 m, floor gap in $[-0.002, 0.002]$ m, tilt ≤ 0.05 rad, top below opening. After release, relative speeds ≤ 0.02 m/s and 0.04 rad/s for 0.10 s, then a 0.50 s dwell. The 0.12 rad gate applies to approach.	200 10 s
t3_phase	Start with the native pre-grasp. Phase alignment: distance ≤ 0.090 m and loop-direction error ≤ 0.20 rad. Enter within 0.030 m at progress ≤ 0.15; reach historical maximum progress ≥ 0.60, then ≥ 0.85 and endpoint distance ≤ 0.055 m.	420 21 s
t3_full	Acquire the wand with bilateral contact for 0.05 s, then complete the phase stages above. In a later sample, retain current progress ≥ 0.85 and move at least 0.045 m from both path and endpoint. Finally open (< -0.5) with no fingerpad contact. Releasing before exit fails.	640 32 s
t4_sphere	Approach within 0.10 m; first qualifying contact after ramp exit; any tray contact before success fails. Bilateral hold for nominal 25 ms (12 physics samples, 24 ms). Raise COM by 0.05 m from its height when the bilateral-hold criterion is first met, and retain contact and height for 0.50 s.	120 6 s
t4_sphere_tabletop; t4_slider	Approach within 0.10 m; contact after ramp exit, while supported, in the capture region, and moving at planar speed ≥ 0.08 m/s. Bilateral hold for 0.05 s, COM lift ≥ 0.05 m from the first attainment of that hold criterion, then contact and height for 0.50 s.	120 6 s
t4_can	Same supported-capture sequence. Before first fingerpad contact, a can-axis absolute dot product with vertical > 0.45 fails as unstable tipping. The configuration does not require a transverse closing axis.	120 6 s
t5_replan	Exactly one selected-side impact group completes before robot contact. At 30 ms after the physical impact-end timestamp, branch-signed lateral speed ≥ 0.035 m/s and yaw rate ≥ 0.35 rad/s. Approach within 0.10 m; bilateral hold with grasp-axis error $\leq 18^\circ$ for 0.05 s; COM lift ≥ 0.05 m from the first attainment of that hold criterion and stable hold for 0.50 s. Impact-group lag is 12 ms and maximum contact is 0.10 s.	120 6 s

fails as stopping/deflecting only. T5 checks invalid state, unsafe contact, early robot contact, wrong striker, extra impact/striker trap, insufficient outgoing change, and boundary events before success. At the horizon, zero completed impacts yields missing selected impact; otherwise the result is timeout.

A.2. Scenario Sampling and Replay

The historical E7/agentview recipes specify proposal ranges (Table S2). Continuous min–max fields use uniform draws from NumPy PCG64; discrete choices are equiprobable before physical-feasibility checks. Evaluation loads each accepted scene's complete stored ScenarioSpec and does not resample its parameters. For harmonic translation,

$$p(t) = c + (0, v_y, 0)t + (A_x \sin(2\pi f_x t + \phi), 0, 0),$$

where $\phi \sim U[-\pi, \pi]$. Thus c differs from the initial position by the phase-dependent offset. T4 targets start at zero linear/angular velocity and evolve under free dynamics; no constant target speed is prescribed. Their slope is continuously sampled from $[5, 15]^\circ$. Ramp-local offsets use the ramp center as origin and lateral, downhill, and normal axes: $p_{\text{world}} = c_{\text{ramp}} + R_x(-\alpha)p_{\text{offset}}$. The center is derived from the exit lip and ramp half-size; offsets are not added directly to the lip. T2-SE3 training uses $\omega_z \sim U[-.25, .25]$; validation excludes $(-.08, .08)$. The other thirteen tasks share train/validation proposal recipes.

Occlusion acceptance requires at least eight initially visible frames with visible fraction $\geq .95$, ten to sixty fully occluded frames (fraction $\leq .05$, lasting 0.5–3.0 s), and at least three reappearance frames with fraction $\geq .8$. These are acceptance criteria, not independent sampling variables.

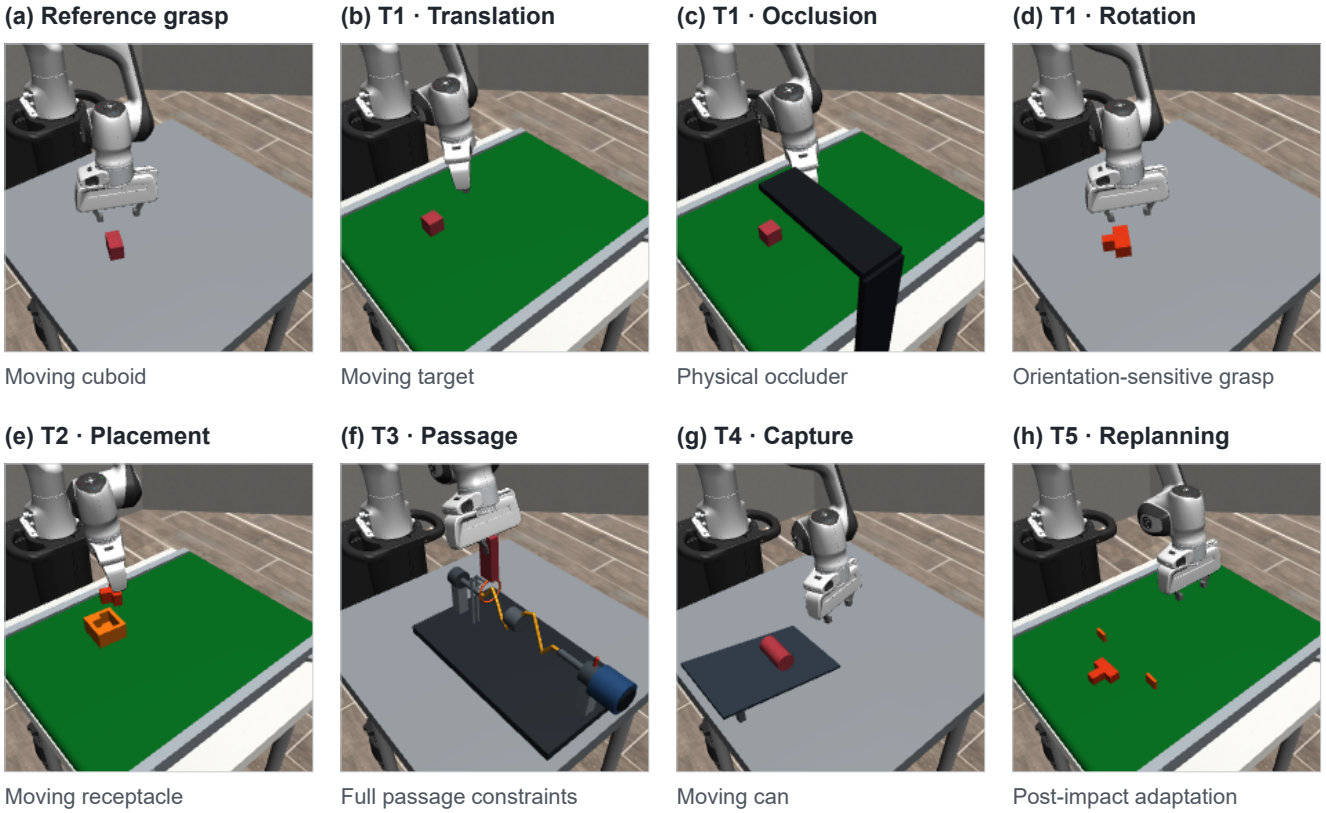


Figure S1. Recorded benchmark observations. Eight examples cover reference grasping; T1 translation, physical occlusion, and orientation-sensitive grasping; T2 moving-receptacle placement; T3 constrained passage; T4 capture; and T5 post-impact adaptation. For each displayed task, the lowest state index in the accepted DEV84 manifest supplies the final observed, uncropped 224×224 agentview RGB. Labels identify task conditions; the snapshots illustrate appearance and geometry, while task outcomes are reported separately. Table S1 lists all fourteen tasks.

For example, the accepted OCC scenes have belt speeds 0.059588–0.099852 m/s within the wider proposal interval. Accepted source seeds can have gaps; a consecutive seed range does not reconstruct the scene set.

All 1,400 stored scenarios share the robot joint initialization

$$q_0 = [0,\ .19634954,\ 0,\ -2.61799388,\ 0,\ 2.94159265,\ .78539816]\ \text{rad},$$

and finger positions $(.020833, -.020833)$ m. T4 additionally stores a temporary preparation-target pose with $x, y \sim U[-.03, .03]$ and yaw $U[0, 2\pi]$, distinct from the release state.

The evaluator source stores recipes in `configs/scenarios/` and task bindings in `configs/tasks/`; the generation entry point is `se3_wam_benchmark/cli/generate_scenarios.py`. The evaluation payload contains the concrete `evaluation_reference/scenes/validation.jsonl` and fourteen `evaluation_reference/scenes/suites/sr_<task>.yaml` files. Each suite uses one repeat and filters the shared manifest by task. These files, with the stored scenario IDs and source seeds, define the replay set used for the paired comparisons.

The accompanying local material package at `results/replay_materials/` contains the archived evaluator, task and sampling configurations, all accepted scenarios, and both ordered selections. Portable copies change only execution paths; originals are retained. The archived manifest parser reads all 1,400 scenarios without changing their scientific fields, and the task/configuration loaders read all fourteen definitions. The supplied checker also verifies disjoint partitions, identities, seeds, and asset references. These checks parse the materials; physics and policy replay require the additional runtime and model assets listed in the package instructions.

A.3. Partitions and Statistical Interpretation

The 140-scene development set has ten scenes per task. Its disjoint complement has ninety per task, totaling 1,260. The future screen reports the two partitions separately. The system-reference study also keeps them separate. The com-

Table S2. Historical validation proposal ranges. Length, linear velocity, angle, and angular velocity use m, m/s, rad, and rad/s unless marked otherwise. Positions are in world coordinates except the stated ramp-local offsets. Continuous intervals are uniform proposals; discrete choices are equiprobable before feasibility checks. Evaluation replays the stored accepted scenarios, whose distribution need not remain uniform. Harmonic centers are not initial positions.

Task	Initial-state proposal	Motion or event parameters
grasp	Harmonic center $c_x \in [-.08, .08]$, $c_y \in [-.26, -.19]$, $z = .822$; yaw $[-1.1, 1.1]$.	$v_y \in [.025, .125]$, $A_x \in [.015, .035]$, $f_x \in [.3, .7]$ Hz; $\omega_z \in [-1.2, 1.2]$.
t1_belt	Target $x \in [-.08, .08]$, $y \in [-.26, -.18]$, $z = .822$; identity orientation, zero initial twist.	Belt $v_y \in [.04, .10]$.
t1_occ	Same target proposal as Belt; canopy at $(.322, -.29, 1.20)$.	Same belt range; clearance parameter .20; physical occlusion.
t1_xyz	Harmonic center $c_x \in [-.025, .025]$, $c_y \in [-.24, -.20]$, $z = .818$; yaw $[-.35, .35]$.	$v_y \in [.04, .09]$, $A_x \in [0, .03]$, $f_x \in [.25, .65]$ Hz; $\omega_z = 0$.
t1_so3	Same as XYZ.	Same translation; $\omega_z \in [-1.2, -.4] \cup [.4, 1.2]$, equal sign probabilities.
t2_trans	Goal $x \in [-.13, -.08]$, $y \in [-.18, -.12]$, $z = .825$; fixed orientation.	Goal $v_y \in [.03, .06]$; no rotation.
t2_se3	Same goal x, y range, $z = .804$; yaw $[-.3, .3]$.	Goal $v_y \in [.03, .06]$; $\omega_z \in [-.25, -.08] \cup [.08, .25]$, equal sign probabilities.
t3_phase	Wand $x, y \in [-.03, .03]$, $z = .870$, yaw $[0, 2\pi]$; wire phase $[-.25, .35]$.	Wire center $(.10, -.02, .94)$; world-X rotation $\omega_x = -\{2.5, 3, \ldots, 7.5\}^\circ$/s (11 equal-probability values).
t3_full	Wand at $(-.05, -.02, 1.04)$ with identity orientation; same wire phase range.	Same wire rotation; cradle and release requirement enabled.
t4_sphere	Ramp-local offsets: lateral $[-.015, .015]$, longitudinal/normal $(-.145, .030)$; lip $(.06, -.02, 1.10)$.	Gravity-driven from rest; gate release at tick 0; slope $[5, 15]^\circ$; target/ramp sliding friction $[.7, .9]$; unsupported capture.
t4_sphere_tabletop	Lateral $[-.015, .015]$, longitudinal/normal $(-.124, .034)$; lip $(0, -.06, .808)$.	Same slope and release; target/ramp friction $[.7, .9]$, table friction .35.
t4_slider	Lateral $[-.015, .015]$, longitudinal/normal $(-.129, .024)$; supported ramp.	Same slope and release; target/ramp friction $[.025, .045]$, table friction $[.012, .018]$.
t4_can	Lateral $[-.010, .010]$, ramp-axis yaw $[-4, 4]^\circ$; longitudinal position follows gate geometry and oriented can extent.	Same slope and release; target/ramp friction $[.25, .45]$, table friction $[.09, .15]$.
t5_replan	$x \in [-.005, .005]$, $y = -.29$, $z = .812$; yaw $[-.08, .08]$; zero initial twist.	Belt $v_y \in [.12, .14]$; equal-probability left/right striker triggered at $y = -.13$; stroke .098, pulse .35 s, actuator scale $[1, 1.1]$, force cap 4 N; command delay 50 ms.

bination screen explicitly selects on all 1,400 scenes while retaining both component counts. Thus its selection score includes development data, and the 1,260 complement is now part of the project's development evidence. Neither membership disjointness nor a different rollout seed makes it a final blind test.

The future study uses each condition's own fourth-epoch endpoint and one training seed. Across arms, the scene, source seed, and inference seed are paired. There are 18,900 accepted outcomes, 1,260 per arm; missing scenes and infrastructure-error exclusions are zero. Inference seeds span 20261058–20262317 for the designated selection indices. Matching an inference seed specifies rollout randomness but does not imply that unlike training graphs consumed identical random-number sequences.

For two policies on the same scenes, let b count candidate successes/reference failures and c candidate failures/reference successes. Then

$$\Delta\mathrm{SR} = 100(b - c)/1260 \tag{S1}$$

in percentage points. The report uses the exact two-sided McNemar test on discordant pairs. Fourteen comparisons to P00 and thirteen to P03 form the twenty-seven-comparison family; adjusted values use $\min(1, 27p)$. The 378 task-level comparisons remain exploratory.

We additionally compute 95% pointwise percentile intervals by resampling paired scenes within each task. For every replicate, each task retains ninety pairs (one hundred in the combination study), and the fourteen task differences are averaged with equal weights. Sampling a multinomial

from each task's four paired outcome counts is equivalent to resampling its binary scene pairs. We use 100,000 replicates and the 2.5th/97.5th percentiles with linear interpolation. The generator is NumPy 2.3.5 PCG64; seeds are 20261017 plus the zero-based lexicographic comparison index for the twenty-seven future contrasts, and 20261017/20261018 for I versus F/E. The post-hoc I–E analysis excluding phase passage retains thirteen tasks with seed 20261019. Tasks are processed in lexicographic order. These intervals condition on the trained checkpoints and fixed task mixture; they do not include training-seed variation or candidate-selection uncertainty. They are not simultaneous intervals; the corrected McNemar family above remains the basis for the overall future-screen comparisons.

A.4. Independent Confirmation and Uncertainty

The 140-scene set and its 1,260-scene complement jointly form the development pool for model and recipe selection. Independent ID confirmation draws 100 new accepted scenarios per task, using the same generator and physical ranges but source seeds absent from all training, validation, and development episodes. The first 100 feasible new scenes in increasing source-seed order define each task's test set. Every method uses these same 1,400 scenarios. The generator, acceptance checks, and partitions are fixed before testing; test outcomes do not select a checkpoint or configuration. Domain-shift conditions and delay sweeps retain their own labels and do not replace ID.

The selected complete configuration, matched action-only control, and strongest development alternative use training seeds 0, 1, and 2. Per-seed scores accompany the task-balanced mean. Paired uncertainty uses 100,000 crossed bootstrap replicates (PCG64 seed 20261018): resample training seed indices jointly across methods, then scene identities within each task jointly across methods and all selected seeds. Average scenes, seeds, and tasks in that order and take the 2.5/97.5 percentiles. The same scene under different seeds is not treated as an independent scene. Intervals are pointwise, quantifying the named contrast. The selection margins and seed-extension rule are given in Appendix E. Existing single-seed tables retain their original conditional uncertainty.

A.5. Inputs and Time

The completed future screen uses one current fixed-camera RGB view, exact depth/camera geometry, robot-state history, and privileged object information. A simulator field stored beside an image is not automatically a deployable sensor observation. Future target states and label-validity masks supervise their respective branches and are excluded from the action condition. A current-object-pose oracle and a future oracle are distinct information conditions.

Physical high-level steps are separated by 50 ms. All one hundred T5 scenarios use a one-step command delay (50 ms); the other thirteen tasks use zero. The native action queue returns a zero command until the delay elapses, then applies the oldest queued command. Forecast labels cover 50–400 ms relative to the source observation. Synchronous reference evaluation pauses physics for computation. The fixed-delay sweep advances physics during the queue delay, and physical deployment records actual capture-to-application age (Appendix F). Future labels follow the demonstration's recorded geometry under its executed actions. The current validity masks check geometric support, projection, and visibility. They do not establish that the same post-contact future remains valid for an alternative candidate action. The generated-future interface and action-conditioned diagnostics are defined in Appendix E.2.

B. Model Interfaces and Reference Budgets

B.1. Representation-Screen Configuration

The reference D model uses `se3wam_v2/split_ffn`, four transformer layers, width 768, eight attention heads, and factorized attention. DINOv3 ViT-L/16 is frozen [11]. A 224×224 agent-view image produces 196 current visual tokens. H0V has zero visual-history query tokens. The information track also supplies exact depth and camera geometry. Projected image patches receive a view embedding, a linear embedding of mean valid metric depth, and a linear embedding of the patch-center robot-base XYZ before the view interface. All reported internal conditions use one agent view, so this interface returns that view directly and does not perform cross-view attention. XYZ is backprojected from fixed source pixels (indices 7, 23, ..., 215) using camera intrinsics and extrinsics. These coordinates also enter 3-D RoPE for visual Q/K and the corresponding flow queries; object, robot, meta, and action tokens receive no XYZ. Robot input retains eight EEF7 states at 20 Hz. Eight object tokens encode current GT pose and seven motion intervals using 25 numeric statistics and pose/interval/acceleration validity flags.

The 25 object dimensions are $(p, \log R, \Delta p, \log \Delta R, \Delta t, v, \omega, a, \alpha)$, with dimensions $(3, 3, 3, 3, 1, 3, 3, 3, 3)$. The eight frames span -350 to 0 ms. Slots 0–6 encode adjacent intervals at their centers (-325 to -25 ms); only slot 7 retains the current absolute pose. Velocity divides interval displacement by elapsed time, and acceleration differences adjacent velocities at their interval centers. No state preceding the window is used. Each dimension is standardized by training statistics and invalid dimensions are zeroed. Three validity flags are appended before a $28 \to 768 \to 768$ MLP with GELU and a time embedding. Eight object, eight EEF, and 196 visual tokens form the scene bank. Sixteen learned meta tokens are separate inputs. The object-motion values come from

simulator truth in this track. The current-pose history track removes historical object poses and all derived interval kinematics while preserving the current pose and robot history. This is a different information condition, not a different name for the same input.

D shares attention QKV/output projections and splits feed-forward networks by world/meta, flow, and action stream. Scene tokens access scene tokens; meta tokens access scene and meta; flow accesses scene and flow; action accesses meta and action. Consequently, flow supervision can change shared weights without an online flow-to-action path. The deterministic pose head pools sixteen final meta tokens.

B.2. Coordinates, Losses, and Generation

Target future translation is cumulative displacement in the fixed metric frame. Rotation is the left-relative vector $\phi_h = \mathrm{Log}(R_{t+\delta_h} R_t^\top)^\vee$. For valid targets $\mathcal{V}$, the pose loss is

$$\mathcal{L}_{\text{pose}} = \frac{1}{|\mathcal{V}|} \sum_{h \in \mathcal{V}} \big[\|\hat{d}_h - d_h\|_2^2 + 0.1\, d_{\mathrm{SO}(3)}(\hat{R}_{t+\delta_h}, R_{t+\delta_h})^2 \big]. \tag{S2}$$

The external weight multiplying this loss is separate from the internal rotation coefficient. A full twist exponential would couple translation to rotation through its Jacobian and is not the implemented parameterization.

The action has eight E7 steps: base-frame translation increment, left-relative rotation vector, and gripper command. Translation and rotation training bounds are 0.05 m and 0.5 rad per axis. The inherited evaluation translation limit is 0.10 m per axis. Training normalization uses the mean and population standard deviation of valid, non-repeated issued E7 commands in the training split, accumulated in FP64 and stored in FP32. Action flow matching uses a unit Gaussian in these coordinates. Each query samples $u \sim U[0, 1)$, shared across its full 8×7 action chunk. Deployment reverses normalization, decodes rotation vectors, and clips physical translation/rotation at 0.10 m/0.5 rad per axis; the gripper command uses its sign. Scene flow uses a metric Gaussian source with standard deviation 0.02 m. Deterministic auxiliary readouts, including pose, do not consume flow-matching source noise and time.

Only the action generator runs during Q2 deployment, using five Euler steps. Conditioning caching is on; temporal/velocity caching is off. Each replan executes one command. Trainable initialization is fresh for seed 0 rather than inherited from a trained common-reference checkpoint. Global batch is 16. Four epochs produce 40,104 updates and 641,664 chunks. AdamW uses learning rate 2×10^{-4}, betas $(0.9, 0.95)$, $\epsilon = 10^{-8}$, weight decay 0.05, and gradient-norm clipping at 1. After 1,000 warm-up updates, cosine decay spans the four-epoch schedule with a minimum learning-rate ratio of 0.1. Training uses FP32 with TF32 enabled, zero dropout, and an FFN expansion ratio of four. P03 weights action, scene flow, and pose losses by one; Alpha is disabled. For cumulative scene flow F_h, the dynamic weight is $w_h = 1 + 128\,\mathrm{clip}(\|F_h - F_{h-1}\|_2/c, 0, 1)$, where $F_0 = 0$ and $c = 0.014620669186115265$ m. The weighted FM loss divides by three times the valid correspondence count, rather than by the sum of weights. P00, P02, and P05 finish with an eight-device topology; other arms finish with sixteen. Some retained runs have recorded topology transitions. All endpoints retain the declared exposure; topology transitions can change the subsequent numerical training trajectory.

B.3. Alternative Target Definitions

Table S3 specifies the target arrays after omitting batch and query axes. Unless stated otherwise, the eight horizons are 50, 100, . . . , 400 ms. Surface correspondence follows fixed source geometry; a future occlusion does not by itself invalidate that identity. Track visibility is a separate label, using depth consistency within $\max(0.001\text{ m}, 0.001z)$ in the camera frame.

Wan receives nine RGB frames at 192×160 normalized to $[-1, 1]$. Its first temporal latent is dropped; the remaining two encode frames 1–4 and 5–8. Codec labels are extracted from actual decoder motion vectors. Target-track sampling starts nearest the mask centroid, then selects the farthest point; row-major order resolves ties. Fewer than 32 points are zero-padded and masked. P08/P09 adapt mechanisms rather than reproduce an entire published model.

Regression objectives average squared error over channels and valid positions. Tracks add mean two-coordinate MSE and visibility binary cross-entropy with logits on valid correspondences. Valid positions are pooled across the batch, rather than averaging episode means. Native pose uses the geometric loss above. Replacement losses P04–P09, native flow, native pose, and action each have outer weight one.

Aalpha adds motion-magnitude MSE with weight 0.5 while preserving P03's Alpha-head initialization. P12, P13, and P14 add future depth, spatial DINO, and dynamic-region RGB supervision, each with weight 0.1. P15 combines the three losses at these same weights. A non-existent head has an inapplicable readout, not a zero error. The combination cannot identify every pairwise interaction. Raw total losses across these objectives are not commensurate.

B.4. System-Reference Training

ACT and Diffusion Policy train from scratch for 50,000 updates at global batch 32. The $\pi_{0.5}$ adaptation starts from `pi05_base` and trains 30,000 updates at global batch 256 [9]. DynamicVLA uses 50,000 updates at global batch 44, initialized from `hzxie/dynamic-vla-DOM` with vision, text, and connector frozen. It uses observations at indices $[-2, 0]$, a 20-step horizon, and normalization fitted on 800 demonstrations. OpenWAM uses a fresh fine-tuning run

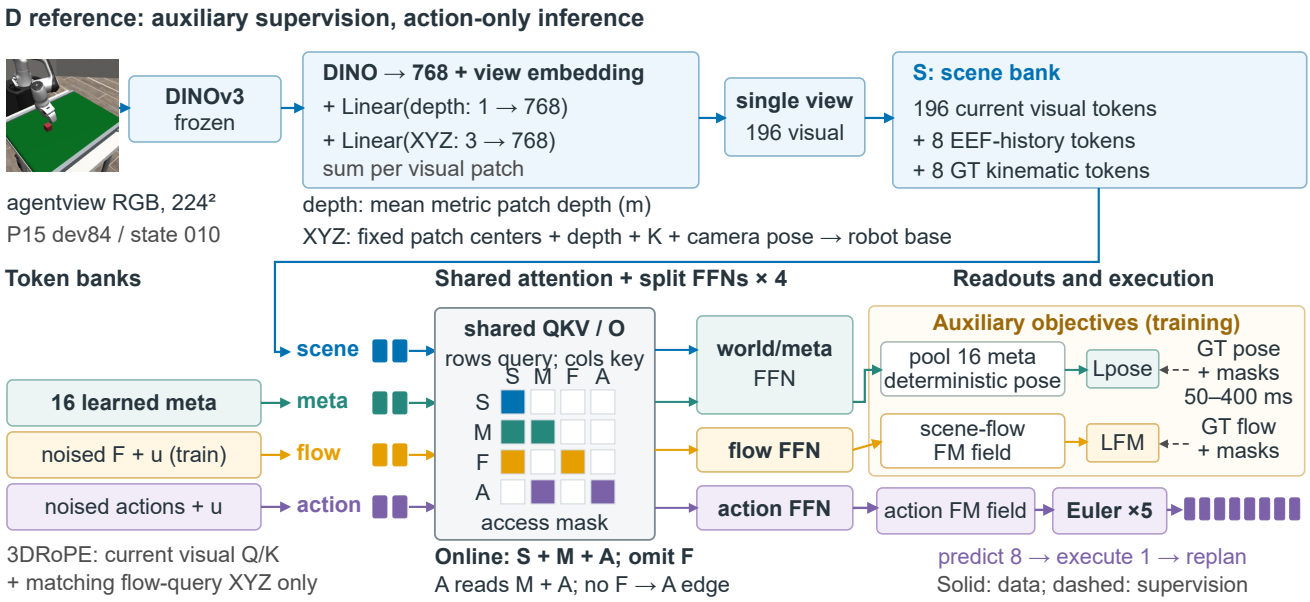


Figure S2. Reference D/P03 computation used in the representation screen. Observation/meta, scene-flow, and Action streams share attention projections with separate feed-forward networks. Scene-flow and target-pose objectives supervise the world representation; Action reads meta context but not the generated future. This graph specifies the auxiliary-supervision reference. Appendix E defines the controlled online-access variants.

Table S3. Future targets in the representation screen. Shape excludes batch/query axes. Displacements are cumulative from the current state except codec vectors, which retain the decoder's native correspondence. All targets come from the recorded demonstration.

Target	Shape	Support, units, and validity
Scene flow	$196 \times 8 \times 3$	Fixed 14×14 patch centers; base-frame meters; valid source depth/geometry and finite future displacement.
Target pose	8×6	Cumulative translation (m) and left-relative rotvec (rad); valid object pose and query.
P04: Wan	$48 \times 2 \times 12 \times 10$	Frozen Wan2.2 VAE posterior mean and native normalization; future groups end at 200/400 ms; valid query.
P05: dense 2-D	$8 \times 2 \times 224 \times 224$	Fixed source pixels; displacement divided by 224; valid source and finite future projection.
P06: codec	$8 \times 2 \times 64 \times 64$	Each future/current pair is encoded as P/I frames; native vectors divided by 64; actual valid vector regions.
P07: scene tracks	$196 \times 8 \times 3$	Same patch source points; two displacement channels divided by 224 and one visibility logit; valid correspondences.
P08: object DINO	8×1024	Object-mask-area-weighted patch pooling; channel layer normalization; positive mask area.
P09: object tracks	$32 \times 8 \times 3$	Deterministic farthest-point sampling in the current target mask; displacement and visibility as P07; padded points invalid.
A_α: magnitude	196×8	Cumulative flow norm (m); same correspondence mask as flow.
P12: depth	$8 \times 1 \times 224 \times 224$	Metric future depth; finite positive pixels.
P13: DINO	$8 \times 196 \times 1024$	Frozen DINOv3-L/16 patches, channel layer normalization; valid query/horizon/patch.
P14: dynamic RGB	$8 \times 3 \times 224 \times 224$	RGB divided by 255; visible surfaces moving more than 1 mm in world coordinates, including robot and other geometry.

from the official Alpha foundation, for five epochs, 34,490 updates at global batch 192, with its declared foundation components. DynamicWAM uses its official full initialization and 40,000 updates at global batch 128 [7, 13]. DreamVLA's 20-epoch and 40-epoch schedules use independent training runs, 48,220 and 96,440 updates at global batch 512; the shorter endpoint is not the midpoint checkpoint of the longer schedule [16]. DreamVLA receives seven causal RGB frames, EEF position/Euler orientation, two finger positions, and fixed task text. Its missing wrist view is black; inference uses DDIM10 and classifier-free guidance 1.5. OpenWAM receives current RGB, seven joint values,

Table S4. System references on the 1,260-scene complement (fourteen tasks, ninety scenes each). Training seed is 0. Methods retain their adapted input, initialization, and execution pipelines; budgets and fixed140 counts are in Appendix B. Scores compare complete systems, not matched architectures. The lower block receives GT target pose and interval kinematics.

System	Initialization / study condition	Successes /1,260	SR (%)
Adapted published systems			
ACT [17]	From scratch	424	33.65
Diffusion Policy [1]	From scratch	432	34.29
$\pi_{0.5}$ [9]	Official `pi05_base`	751	59.60
DynamicVLA [15]	Adapted native pipeline	363	28.81
OpenWAM-Alpha [13]	Official Alpha foundation	708	56.19
DynamicWAM [7]	Official full initialization	670	53.17
DreamVLA [16]: 20 epochs	Independent 20-epoch training schedule	1,000	79.37
40 epochs	Independent 40-epoch training schedule	1,003	79.60
Controlled-study systems with privileged target information			
P00: no future supervision	Current image; frozen DINO-L	791	62.78
P03: scene flow + pose	Current image; frozen DINO-L	786	62.38
I: scene tracks + pose	Full visual window; CNN; target RGB reconstruction	839	66.59

Table S5. Training exposure and development counts for all system-reference endpoints. Batch is global; all endpoints use training seed 0. Initialization and execution are part of each adapted system, not matched factors. The two DreamVLA endpoints are independent training schedules. The 1,260-scene counts appear in the main system table.

System	Updates	Batch	Success /140	Execution / initialization note
ACT	50,000	32	50	Scratch; temporal aggregation
Diffusion Policy	50,000	32	54	Scratch; execute 8 actions
$\pi_{0.5}$	30,000	256	89	`pi05_base`; execute 10
DynamicVLA	50,000	44	24	Official DOM checkpoint; native continuous pipeline
OpenWAM-Alpha	34,490	192	76	Official Alpha foundation; execute 32
DynamicWAM	40,000	128	72	Official full initialization; execute 16
DreamVLA: 20 epochs	48,220	512	112	Independent schedule; aggregate forecasts, execute 1
40 epochs	96,440	512	114	Independent longer schedule
Study P00	40,104	16	88	Fresh task network; frozen DINO-L; execute 1
Study P03	40,104	16	93	Same reference recipe as P00
Study I	40,104	16	96	Fresh CNN condition; full visual history; execute 1

measured gripper opening, and task text. One agent view and two black wrist views form a 384×320 composite; state is mapped to its native 80-dimensional slots. Ten synchronous joint inference steps produce nine video frames and 32 actions, with video decoding disabled.

Executors also differ: ACT uses temporal aggregation, Diffusion Policy executes eight actions, $\pi_{0.5}$ ten, DynamicWAM sixteen, and OpenWAM thirty-two before replanning. DreamVLA predicts three steps but replans every tick, aggregates the latest three forecasts with exponential weights $\exp(-0.01k)$ over their age order, and executes one command. The internal representation screen executes one. System-level interpretation retains these differences. Matched-budget comparisons must also report each system's inputs, initialization, and execution configuration.

C. History Encoding and Conditional Information

C.1. History Conditions

The current-pose study compares eight history encodings plus a fresh GT-history reference. Every visual-history condition preserves current GT target pose and eight robot-state samples, while historical object pose and derived kinematics are masked. Its reference future supervision is held fixed. The eight current-pose conditions are current image, Window, Codec-MV, Flow, Flow12D, PACT-Age, Wan, and PACT-Uniform. The fresh GT-history reference uses the same 1,400-scene evaluation and probe protocol. It retains current RGB and adds privileged motion history. PACT-Age distributes fifty-six history queries from older to newer

frames as $[2, 4, 6, 8, 10, 12, 14]$; PACT-Uniform uses seven groups of eight. These adapt PACT-WAM's temporal allocation principle [6] through per-frame query compression, rather than reproducing its complete model.

All windows contain eight samples at 20 Hz, spanning -350 to 0 ms. Current/Window/PACT add fixed 768-dimensional sinusoidal frame-time embeddings in seconds; time subtraction is performed in FP64 before conversion to FP32. Window retains every past patch. PACT uses one shared eight-query bank; age allocation linearly interpolates that bank with aligned endpoints, rather than adding frame-specific query parameters. Layer-normalized multi-head attention produces query-residual outputs.

Flow is bidirectional Farnebäck flow [3] on area-resized 64×64 RGB (pyramid scale 0.5, three levels, window 15, three iterations, polynomial neighborhood 5, sigma 1.2, flags 0). The four intervals last 100, 100, 100, and 50 ms. Validity requires finite vectors, in-frame endpoints, valid bilinear reverse sampling, and

$$\|f + b(x+f)\|^2 \le 0.01(\|f\|^2 + \|b(x+f)\|^2) + 0.5. \quad \text{(S3)}$$

Flow color uses direction as hue, unit saturation, and magnitude divided by the interval's 99th percentile, clipped to $[0, 1]$, as value. Invalid pixels are black with validity zero. The four-channel field enters a $4 \to 768$ convolution with kernel/stride 8, followed by layer normalization, 64 learned spatial embeddings, and fourteen shared queries with interval-time embeddings. Attention, residual addition, and layer normalization produce fourteen tokens per interval.

For valid flow $f = (d_x, d_y)$, let $r = \|f\|_2$, bars denote pixel means, and q be the 99th percentile of r. Flow12D supplies

$$(\bar{d}_x, \bar{d}_y, \bar{r}, q, \Delta t, \bar{d}_x/\Delta t, \bar{d}_y/\Delta t, \bar{r}/\Delta t, q/\Delta t, a_x, a_y, \|a\|_2).$$

Acceleration differences adjacent mean velocities over interval-center time gaps of 100, 100, and 75 ms; the first interval has no acceleration. All dimensions except Δt are divided by 64, with no fitted mean/standard deviation. Twelve feature-valid flags form a 24-dimensional input to a $24 \to 768 \to 768$ GELU MLP. A separate $2 \to 768 \to 768$ MLP embeds the left/right times relative to the current frame. Pure Flow zeros the numeric values and flags but retains these four slots and their time embeddings. Thus the comparison adds magnitude, velocity, and acceleration statistics to a color encoding that normalizes magnitude.

Codec labels come from independent MPEG-4 Part 2 I/P pairs with no B frames, GOP 2, quantizer 2, YUV420p, one thread, and bitrate 10^6. Negated decoder backward vectors remain attached to destination blocks, unlike source-grid optical flow. Intra, uncovered, and out-of-frame-reference regions are invalid; no forward/backward confidence filter is applied. Wan uses a frozen FP32 Wan2.2 VAE posterior mean after right-padding T8 with the current frame. Its temporal groups end at $-350, -150, 0$ ms, with real-frame fractions $1, 1, 0.75$. A $48 \to 768$ kernel/stride-2 convolution creates 147 source tokens; learned spatial embeddings and an MLP for group time/fraction precede 56-query compression. VAE caches are cleared per window.

The original implementation allocates 1,584 scene slots for Current/Window, 272 for Flow/Flow12D, and 268 for the other compressed conditions. These counts include masked past-object slots; sixteen meta tokens are additional. Allocated slots and valid information are reported separately.

Probe extraction covers 350 episodes and 1,050 windows per condition, with 306 valid readouts. Shared recorded inputs, labels, initial flow-matching noise, and the frozen current-image, geometry, robot-state, and clock baseline agree across conditions. Oracle inputs agree across the eight current-pose conditions; the intended GT-motion difference is excluded from this comparison. No visual-history condition exceeds the current-image control's point estimate. Outcomes are paired by task, scene, source seed, and inference seed. Against Current, gain/loss counts are 137/223 for Window, 133/148 for Codec-MV, 176/204 for Flow, 183/208 for Flow12D, 132/246 for PACT-Age, 194/232 for Wan, 122/200 for PACT-Uniform, and 238/162 for GT history. The main-paper intervals use Appendix A's task-stratified bootstrap with 100 paired scenes per task and 100,000 replicates. Seeds are 20261017–20261024 in this order. They describe scene uncertainty for the E4 seed-0 endpoints on the development pool, without training-seed, selection, or multiplicity adjustment. Outcome pairing and probe-input comparisons cover distinct experimental records.

GT history has 586 joint successes and 414 joint failures with Current; its net gain is 76 scenes, or 5.43 points (95% interval 3.07–7.79). Table S8 shows a pronounced task tradeoff: XYZ grasping gains forty successes, while phase passage falls from 84/100 to 1/100. The aggregate score therefore does not imply a uniform gain from motion input.

C.2. Recipe Differences and Cross-Round Context

This study uses frozen DINO-L, retains the original Alpha objective, and has a cosine schedule over the first 10,026 updates followed by three epochs at 2×10^{-5}. The combination study uses DINO-S or CNN, disables Alpha, and applies cosine scheduling across all 40,104 updates. Its stronger score therefore cannot be subtracted from the history-study score to estimate a visual-history effect. Flow12D and Wan train fresh on sixteen PRO6000 devices; PACT-Uniform trains on eight PRO6000 devices, with a same-topology restart from update 781. Window, Codec-MV, and Flow resume sixteen-device A100 prefixes on PRO6000; PACT-Age additionally changes from sixteen to eight devices. All preserve global batch 16 and the four-epoch exposure. The device histories preclude a matched throughput comparison. GT history

Table S6. History encoders in the current-pose study. All conditions preserve the current 196 DINO-L visual tokens, T8 EEF states, and current object pose. History tokens are width 768 and carry no XYZ. Counts below concern added visual/numeric history tokens, not allocated scene slots.

Condition	Sampled times (ms)	History encoding	Added tokens
Current	0	Past image slots are masked.	0
Window	$-350{:}50{:}0$	Seven past frames, 196 DINO patches each; no query compression.	1,372
Flow	$-350, -250, -150, -50, 0$	Four $2 \times 64 \times 64$ flows; each RGB+validity field yields 64 patches and 14 learned-query outputs.	$56 + 4$
Flow12D	Same as Flow	Same visual encoder; four numeric slots receive twelve motion statistics and their flags.	$56 + 4$
Codec-MV	Same as Flow	Four actual codec MV fields, encoded by the same RGB+validity backend; no numeric slots.	56
Wan	$-350{:}50{:}0$	Repeat current frame once; $48 \times 3 \times 14 \times 14$ VAE latent $\rightarrow$ 147 patches $\rightarrow$ 56 queries.	56
PACT-Uniform	$-350{:}50{:} - 50$	Shared eight-query bank reads each of seven past frames.	56
PACT-Age	Same as Uniform	Shared query bank interpolated to $[2, 4, 6, 8, 10, 12, 14]$ per frame.	56

Table S7. Complete history results. The total is a development summary, not a separate held-out policy test.

Condition	/140	/1,260	/1,400	SR (%)
Current image	80	668	748	53.43
Window	69	593	662	47.29
Codec-MV	77	656	733	52.36
Flow	76	644	720	51.43
Flow12D	75	648	723	51.64
PACT-Age	65	569	634	45.29
Wan	70	640	710	50.71
PACT-Uniform	68	602	670	47.86
Reference with privileged target history				
Fresh GT history	84	740	824	58.86

runs on eight PRO6000 devices until update 18,060, then resumes on sixteen with a native RNG reset. Its completed endpoint has 40,104 updates and 641,664 sampled chunks. This topology change is included in the interpretation of the single-training-seed comparison.

An earlier eight-condition study retained GT motion in every visual encoder (Table S9). Its current-image endpoint scores 835/1,400 (88/140 plus 747/1,260), distinct from the fresh 824/1,400 reference. No visual-history encoder exceeds the corresponding current-image point estimate in either round. Removing GT motion therefore does not reveal an overall visual-history advantage in the tested recipes. Cross-round differences remain descriptive: device, topology, and recovery histories differ, and each condition has one training seed. They do not identify a matched factorial interaction. Repeated current-pose/GT-motion $\times$ current-image/history training would assess its stability and interaction under a common topology.

C.3. Offline History Perturbations

For each of fourteen tasks, the probe protocol assigns 15/5/5 distinct episodes to training/validation/test, with three windows per episode. Table S10 uses the resulting 70 test episodes and 210 windows, with initial-noise seed 17. This is a probe split within the 350 policy-validation episodes, not a new policy test set. The intervention repeats the current image in all seven past slots or reverses the seven past images. Depth and calibration follow the same image permutation; the current frame, object information, robot history, timestamps, and masks remain fixed. These perturbed histories need not be physically realizable.

Each intervention re-encodes the window without a previous perception cache, rebuilds the condition, and reuses the actual Action/Flow initial-noise arrays. The native reports accept 840 same-input/noise controls per condition. Repeated-baseline and identity controls have zero differences; both current-image references have zero action and readout changes under every history intervention. Separate three-step checks in the eight current-pose conditions confirm exact condition-cache reconstruction and zero influence of masked old GT states. The GT-history reference intentionally consumes those states and is excluded from that masking check. Future labels are read only after prediction.

All seven history encodings change the issued first action under these perturbations, despite their lack of a success-rate gain in the main screen. For example, Window translation changes by 9.83 mm under repetition and 1.92 mm under reversal; Wan changes by 20.05 and 24.25 mm. The comparison distinguishes sensitivity to history from task benefit. It does not measure a closed-loop intervention or assign a mechanism to the success-rate differences.

This analysis aggregates the original per-task, episode-averaged reports. Their sample identities and source reports

Table S8. History-study successes, one hundred scenes per task and condition. The first eight conditions retain current GT target pose; GT motion retains current RGB and adds privileged target-motion history. All use one training seed and the fourth-epoch endpoint; the 1,400 scenes are the development pool.

Task	Current	Window	Codec	Flow	Flow12D	PACT-Age	Wan	Uniform	GT motion
Grasp	66	71	62	65	66	62	62	52	88
Belt	87	48	89	99	94	65	88	61	95
Occlusion	72	67	62	74	70	59	87	70	94
SO(3)	42	36	34	38	44	29	37	17	56
XYZ	41	48	54	53	47	55	62	44	81
SE(3) placement	0	0	0	0	0	0	0	0	0
Trans. placement	13	6	15	13	14	12	40	16	10
Full passage	0	0	0	0	0	0	0	1	0
Phase passage	84	6	99	0	0	28	3	67	1
Can	70	96	82	79	92	79	53	68	100
Slider	84	80	70	80	85	59	73	70	78
Sphere	41	55	46	65	76	38	63	45	66
Tabletop sphere	96	97	98	100	100	77	100	96	99
Replan	52	52	22	54	35	71	42	63	56
Total / 1,400	748	662	733	720	723	634	710	670	824

Table S9. Cross-round history context: successes over the same 1,400 scenes. The earlier round gives every encoder GT motion; the current-pose round removes it. Both retain current GT pose and robot history. Training-device and recovery histories differ, so this is not a matched factorial comparison. The earlier current-image endpoint (835) differs from the freshly trained GT-history reference (824).

Visual encoding	Earlier: GT motion	Current pose only
Current image	835	748
Window	834	662
Codec-MV	797	733
Flow	806	720
Flow12D	789	723
PACT-Age	816	634
Wan	782	710
PACT-Uniform	783	670

Table S10. Offline sensitivity in the eight current-pose conditions. Mean change in the issued first action, relative to the unchanged input, over 70 probe-test episodes (210 windows, fourteen tasks). Values are equal-task means of episode means. Translation is L2 change in mm; rotation is geodesic change in degrees. These are action changes, not action errors or success gains.

	Repeat current		Reverse past	
Condition	mm	deg	mm	deg
Current image	0.00	0.00	0.00	0.00
Window	9.83	1.28	1.92	0.34
Codec-MV	10.89	1.59	13.13	2.00
Flow	17.80	2.20	15.15	2.51
Flow12D	16.21	2.43	16.85	2.95
PACT-Age	6.07	0.48	2.17	0.25
Wan	20.05	2.70	24.25	3.98
PACT-Uniform	7.50	0.67	0.33	0.04

match the native records; the separate array audit above verifies the shared inputs and initial noise.

C.4. Frozen Readouts and Conditional Residuals

The readouts test which recorded motion variables can be decoded from frozen features. They use the same 15/5/5 episode split above, selected by sorted native episode identity. Ridge regression is fitted independently for each task, feature site, and label. The penalty is chosen from $\{0.1, 1, 10, 100\}$ by validation coordinate MSE, without refitting on validation or test data. Features use training-only means and population standard deviations, with values at most 10^{-8} replaced by one; targets are centered but not scaled, and the intercept is unpenalized. A shuffled control permutes the forty-five training-window labels with seed 1729, leaving validation and test labels unchanged.

Features and targets. The history site captures the visual-history encoder output entering the first Transformer layer: Window pools spatially within each past frame, while compressed encoders retain ordered tokens. The other sites are final Meta V and the input immediately before the action head at the first Action-branch function evaluation, $\tau = 0$, each mean/max pooled to 1,536 dimensions. The 18,048-dimensional conditioned baseline contains frozen current

DINO-L features pooled to 4×4, current depth and validity, robot-base patch coordinates and validity, eight EEF states, relative times and masks, and the condition's actual oracle values and validity. Thus GT history supplies additional motion information despite equal baseline width. Past-position labels refer to the target body origin in the robot-base frame at lags of 50, 200, and 350 ms. Velocity uses the last 50 ms backward difference; the future label is the demonstrated 400 ms continuation displacement.

Readout comparisons. A retained readout uses site features alone; a joint readout concatenates them with the conditioned baseline. Joint gain is baseline physical error minus joint physical error. For conditional residuals, the baseline penalty is selected on validation data, then five episode-disjoint folds produce training residuals: twelve episodes fit each baseline and three provide its held-out targets. Site features predict these residuals. Validation and test residuals instead use the baseline fitted on all fifteen training episodes. Reconstruction adds this fixed baseline prediction to the predicted residual. This tests a restricted decoder, not all information available to the policy.

Aggregation and uncertainty. Position error is the per-window Euclidean distance, averaged over three windows per episode, five test episodes per task, and fourteen equally weighted tasks. It is not RMSE or the model's pose-head error. Translation and velocity metrics cover all fourteen tasks; rotation and angular velocity cover only the three orientation-observable tasks. Per-task intervals resample the five episode means 2,000 times with seed 1729; they condition on the fitted probes and policy endpoints. Macro results average task point estimates and have no confidence intervals.

Results. Window has lower 350 ms position-error point estimates than Current at both Meta V (8.46 versus 10.89 mm) and the action input (8.31 versus 10.99 mm), while its success rate is 6.14 points lower (Fig. S3). Readable past state and sensitivity to image history therefore coexist with worse task performance in this comparison. The conditioned residuals give a similarly limited result: across three sites, nine conditions, and three labels—350 ms past position, last-50 ms velocity, and 400 ms future displacement—74 of 75 defined task-macro R^2 values are negative; the exception is Flow12D Meta V velocity at 0.034. Residual R^2 uses the test-residual mean as its reference, and its macro averages defined task scores rather than pooling tasks. A negative value does not establish absence of encoded information. Current pose is oracle-determined in every condition; GT history also determines past pose and velocity. Their unique-information residual scores are undefined, not zero, and baseline approximation errors are reported separately. The six GT-history past-position/velocity cells are therefore excluded from the count above. These readouts do not establish that the conditioned input is an information-theoretic sufficient statistic.

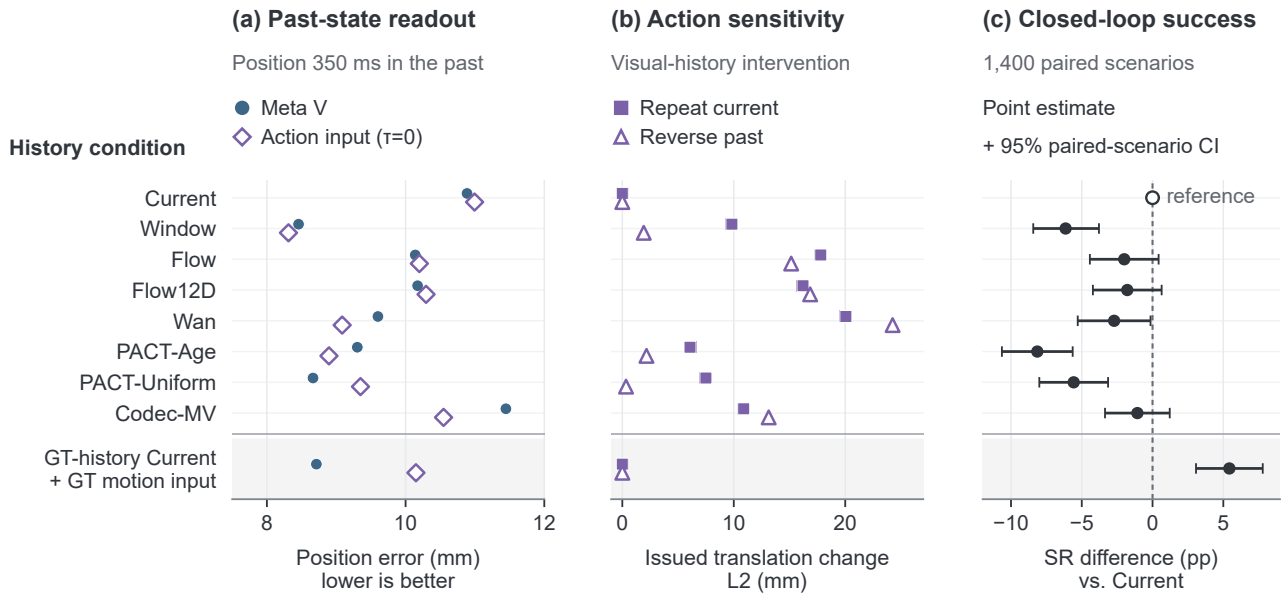
C.5. Frozen Object-Motion History Intervention

The frozen-history intervention fixes the fresh GT-history E4 endpoint and changes only its object-history input at inference. In the native object-descriptor path, every past target pose is replaced by the current pose before the native function recomputes interval kinematics. This *stationary history* removes past displacement, relative rotation, velocity, and acceleration before normalization, up to floating-point round-off. Current target pose, RGB, robot history, timestamps, validity masks, and model weights are unchanged within each call. Later closed-loop observations follow each resulting trajectory; they are not held equal across interventions. The replacement follows the temporal frame cache, and conditioning is rebuilt so cached descriptors cannot restore the original motion.

Controls and sampling. The action test reuses the 70 test episodes and 210 windows of the probe split, with initial-noise seed 17 and the same actual noise arrays for native, identity-copy, and stationary-history calls. Identity copying preserves all 210 action arrays elementwise. Three consecutive online windows from one episode additionally verify full recomputation, incremental temporal caching, and conditioning-cache reconstruction. In closed loop, identity copying reproduces all original fixed140 success/failure labels, including its 84 successes. The follow-up then evaluates stationary history on all 1,400 original scenes, preserving scene identities, source/inference seeds, five Euler steps, and the original execution interface. The true-history 1,400-scene result is reused from the accepted endpoint; no training or additional true-history evaluation is counted.

Action response and closed-loop dependence. All 210 first issued actions change under stationary history. Mean first-translation change is 18.40 mm (95% interval 16.90–19.93), and mean rotational change is 0.0424 rad (0.0368–0.0482). Success falls from 824/1,400 (58.86%) to 9/1,400 (0.64%): 815 originally successful scenes fail and no originally failed scene succeeds. The paired drop is 58.21 points (95% interval 56.50–59.86). The remaining successes are seven reference-grasp scenes and two orientation-sensitive-grasp scenes. Rigid-motion placement and full passage remain unsuccessful in both conditions. Figure S4 shows the action distribution and all fourteen tasks.

The follow-up retains its native 20,000-replicate bootstrap with seed 1729, separately from the main history table's 100,000-replicate analysis. Closed-loop intervals resample paired scenes within each of fourteen fixed tasks. Action metrics first average the three windows of each episode,



Seed 0. Offline: 14-task means, no CIs. Closed loop: scenario uncertainty only.

Figure S3. History decodability, action sensitivity, and closed-loop success. (a) Frozen ridge error for position 350 ms in the past, from final Meta V or the first action-head input ($\tau = 0$). (b) L2 change of the first issued translation after action limits, under repeat-current or reverse-past intervention with identical initial noise. Panels (a,b) show descriptive means across fourteen fixed tasks, without confidence intervals. (c) Success-rate difference from Current over 1,400 paired scenarios, with pointwise 95% intervals from 100,000 task-stratified resamples. These intervals condition on single-seed checkpoints and the development pool. The separated GT-history reference receives additional GT motion. Both current-image references have no visual history to perturb; GT motion remains intact. Lower readout error and nonzero action sensitivity do not establish a task benefit.

then resample the five test episodes within each task and average tasks equally. Thus the action interval has seventy episode units, not 210 independent windows. Both intervals condition on one frozen endpoint and exclude training-seed uncertainty.

These controls establish that this endpoint uses motion history in action generation and closed-loop control, and that the learned policy is sensitive to its removal at inference. The 58.21-point loss measures this input mismatch; Current trained without GT motion still achieves 53.43%. Together with the training comparison and visual-history probes, this supports retaining structured motion history in the reference recipe. It does not establish SE(3) sufficiency, mediation of all visual history through SE(3), or a compute advantage. Translation/rotation contributions and robustness to estimated, delayed motion histories remain separate questions.

C.6. Matched-Current, Visibility, and Sampling Tests

The complementary history tests match task, object identity, camera, target pose within 5 mm/5°, and current EEF pose within 5 mm/5°. Among valid development histories, the nearest different episode supplies the alternate prefix. Current observation, target pose, robot history, and initial action noise stay fixed. Matching coverage and residual pose differences accompany the result; the stationary-history intervention above remains a separate deliberately mismatched input test. The direction-ambiguous subset additionally requires a negative dot product between the two valid last-interval target velocities, each with magnitude at least 0.01 m/s. Subset membership is determined without policy outcomes.

Visibility tests use physical occluders and accepted occlusion durations of 0.5, 1, 2, and 3 s, together with an unobstructed control; current mask/pose validity follows each observation. Sampling tests retain a common 350 ms span and select 2, 4, or 8 actual frames at the nearest available evenly spaced timestamps, always including the current frame. Their time embeddings use actual offsets. A separate observation-age sweep adds 0, 50, 100, and 200 ms before the unchanged history interface. These tests use seeds 0–2 and common confirmation scenes, with natural task eligibility stated for each stratum. Reported quantities are SR, first-issued action change, conditional motion-readout error,

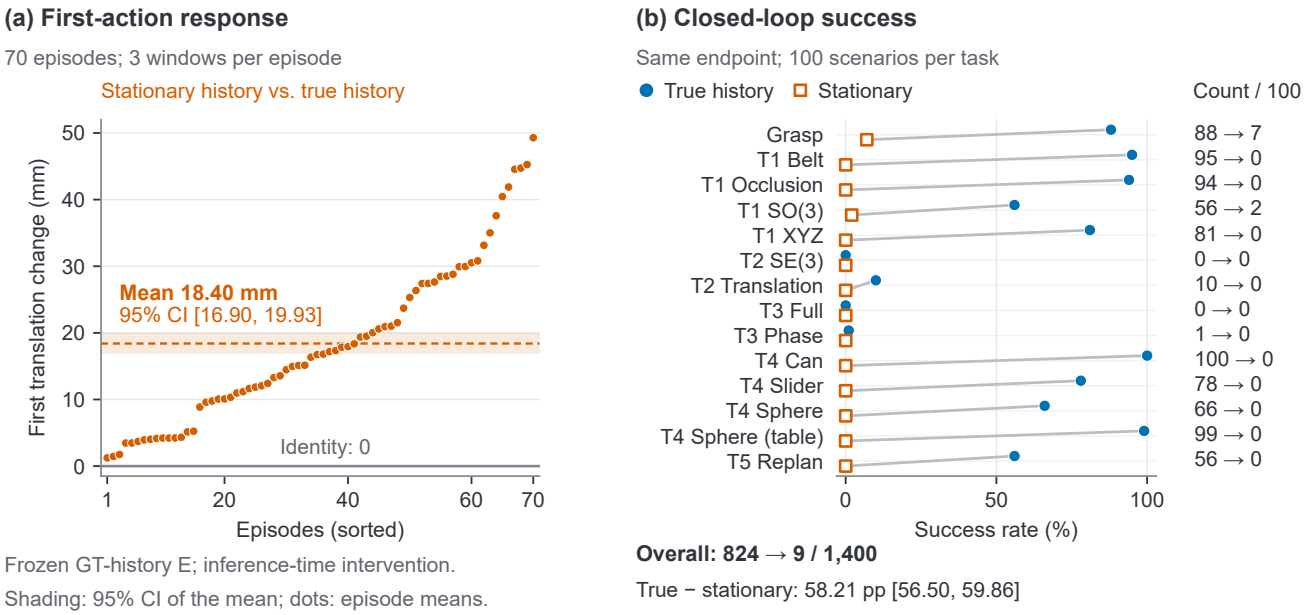


Figure S4. A frozen policy depends on correct object-motion history. Stationary history repeats the current target pose in past slots and recomputes kinematics; other input paths are retained within each call. (a) L2 changes in the first issued translation after action limits. Each point averages three probe windows from one of seventy episodes; the mean and 95% interval use task-stratified episode resampling. Identity copying changes no action. (b) True versus stationary history on fourteen tasks with 100 paired scenes each. The aggregate paired drop uses task-stratified scene resampling. Both intervals use 20,000 replicates, seed 1729. True-history outcomes are reused from the original endpoint; identity copying separately reproduces its fixed140 success/failure labels. This inference-time mismatch effect is distinct from the 5.43-point training-comparison gain of adding GT motion.

and contact time/relative-pose error. Non-contact failures remain in the SR denominator and have an absent, rather than zero, contact error.

D. Future Supervision and Combination Selection

D.1. All Fifteen Future Conditions

All conditions in Table S11 have action supervision. Scene-flow coverage includes target regions, so P01 is not a pure context-only condition. P03 versus P01/P02 compares target-pose and scene-flow objectives under the common action path; it does not isolate an exclusive target/context spatial partition. Main-paper Fig. 3 visualizes all fifteen point estimates with task-stratified paired bootstrap intervals, as specified in Appendix A.

P00/P01/P02/P03 test no future objective, scene flow, pose, and their combination. P04–P07 replace the scene target while retaining pose. P08/P09 replace the object target while retaining scene flow. Aalpha and P12–P15 add supervision to P03. Full target definitions and source units are given in Appendix B. None of these conditions changes deployment into online forecast consumption.

D.2. Paired Outcomes and Task Heterogeneity

Table S12 gives gains, losses, difference intervals, and adjusted exact McNemar probabilities against both reference conditions. P05's adjusted probabilities are 0.00844 versus P00 and 0.01877 versus P03. The remaining positive overall differences are not established advantages. The absence of a detected difference is not an equivalence test.

Table S13 gives every task count with equal denominators. The strongest task-specific changes are exploratory. For P05 versus P03, rolling-sphere success falls from 50 to 23 while `tray_contact` failures rise from 40 to 67. Conversely, post-impact replanning rises from 54 to 60. P09's gains in translational placement and replanning coexist with fewer orientation-sensitive grasps. These observations narrow the conditions to investigate, without treating a native failure code as a learned-mechanism explanation.

The lowest count of native safety failures is P06's 250, but its 96 timeouts are the highest. P15 has only 37 timeouts and 343 safety failures. A favorable change in one category must be interpreted alongside all outcomes. Every policy failure remains in the 1,260-scene denominator.

Table S11. All fifteen future-supervision conditions, grouped by the objective changed. Each uses one training seed, four epochs, common privileged object-motion input, and action-only deployment. The 140 development scenes and 1,260-scene complement remain separate. Scene flow includes target regions.

		Development	Disjoint complement	
ID	Future supervision	Success / 140	Success / 1,260	SR (%)
Reference and component objectives				
P00	No future supervision	88	791	62.78
P01	3-D scene flow	95	782	62.06
P02	Target pose	92	789	62.62
P03	Scene flow + target pose	93	786	62.38
Scene replacements: target-pose objective retained				
P04	Wan latent + pose	92	783	62.14
P05	Dense 2-D displacement + pose	87	746	59.21
P06	Codec motion + pose	86	795	63.10
P07	Sparse scene tracks + pose	90	807	64.05
Target replacements: scene-flow objective retained				
P08	Scene flow + object DINO	89	794	63.02
P09	Scene flow + target tracks	88	808	64.13
Additional objectives: scene flow + target pose retained				
A_α	P03 + motion magnitude	89	778	61.75
P12	P03 + depth	90	791	62.78
P13	P03 + DINO	92	788	62.54
P14	P03 + dynamic RGB	89	790	62.70
P15	P03 + depth / DINO / dynamic RGB	94	802	63.65

Table S12. Paired outcomes over the same 1,260 scenes. Gain means candidate success/reference failure; loss means candidate failure/reference success. Intervals are pointwise task-stratified paired bootstrap intervals for the SR difference, conditional on the trained checkpoints (Appendix A). Exact two-sided McNemar tests use the full family of fourteen P00 and thirteen P03 contrasts: $p_{\mathrm{adj}} = \min(1, 27p)$. P03 against itself is inapplicable.

	P00: no future supervision				P03: scene flow + target pose			
ID	Gain	Loss	95% CI (pp)	Adjusted p	Gain	Loss	95% CI (pp)	Adjusted p
Reference and component objectives								
P01	80	89	$[-2.70, 1.27]$	1	72	76	$[-2.14, 1.51]$	1
P02	61	63	$[-1.83, 1.51]$	1	61	58	$[-1.43, 1.90]$	1
P03	64	69	$[-2.14, 1.35]$	1	—	—	—	—
Scene replacements: target-pose objective retained								
P04	74	82	$[-2.54, 1.27]$	1	71	74	$[-2.06, 1.59]$	1
P05	53	98	$[-5.40, -1.75]$	0.00844	47	87	$[-4.92, -1.43]$	0.0188
P06	81	77	$[-1.59, 2.22]$	1	68	59	$[-1.03, 2.46]$	1
P07	85	69	$[-0.63, 3.17]$	1	81	60	$[-0.16, 3.49]$	1
Target replacements: scene-flow objective retained								
P08	75	72	$[-1.59, 2.06]$	1	59	51	$[-0.95, 2.22]$	1
P09	91	74	$[-0.56, 3.25]$	1	79	57	$[0.00, 3.49]$	1
Additional objectives: scene flow + target pose retained								
A_α	60	73	$[-2.78, 0.71]$	1	53	61	$[-2.30, 1.03]$	1
P12	73	73	$[-1.83, 1.83]$	1	63	58	$[-1.27, 2.06]$	1
P13	70	73	$[-2.06, 1.59]$	1	60	58	$[-1.51, 1.83]$	1
P14	71	72	$[-1.90, 1.75]$	1	62	58	$[-1.35, 2.06]$	1
P15	74	63	$[-0.95, 2.70]$	1	63	47	$[-0.32, 2.94]$	1

Table S13. Future-study successes: ninety scenes per task and 1,260 per condition. Columns follow the task families in Table S1; all tasks are retained, including zero-success tasks. Occl. denotes physical occlusion; Trans. denotes translating-receptacle placement; Tabletop denotes tabletop sphere capture.

	Reference	T1: target motion				T2: placement		T3: passage		T4: capture				T5	
ID	Grasp	Belt	Occl.	SO(3)	XYZ	SE(3)	Trans.	Full	Phase	Can	Slider	Sphere	Tabletop	Replan	Total
Reference and component objectives															
P00	85	90	90	75	76	0	8	3	15	90	69	45	86	59	791
P01	84	90	90	64	69	0	18	0	11	90	74	46	90	56	782
P02	87	90	90	78	73	0	7	0	0	90	71	59	89	55	789
P03	86	90	88	75	77	0	7	0	6	86	80	50	87	54	786
Scene replacements: target-pose objective retained															
P04	83	85	84	73	76	0	20	0	0	90	72	54	89	57	783
P05	84	89	90	66	75	0	10	0	0	90	71	23	88	60	746
P06	86	88	89	79	70	0	6	4	2	89	72	60	85	65	795
P07	82	90	90	78	71	0	17	0	0	89	79	55	90	66	807
Target replacements: scene-flow objective retained															
P08	84	90	90	71	81	0	13	0	0	90	77	60	90	48	794
P09	84	90	89	65	78	0	21	0	0	90	77	59	90	65	808
Additional objectives: scene flow + target pose retained															
A_α	86	90	90	72	75	0	3	0	0	85	73	55	90	59	778
P12	83	90	90	76	76	0	5	0	0	89	77	57	90	58	791
P13	86	90	90	80	73	0	6	0	0	88	78	54	90	53	788
P14	85	89	90	72	79	0	5	0	0	90	80	53	88	59	790
P15	87	90	90	79	81	0	11	0	0	90	75	54	90	55	802

Table S14. Native evaluator failure counts out of 1,260 scenes per condition, in the same design groups as the success tables. Safety is the evaluator category, not a certified safety measure. Every failure remains in the success-rate denominator.

ID	Future supervision	Boundary	Safety	Task viol.	Timeout	Total
Reference and component objectives						
P00	No future supervision	84	297	17	71	469
P01	3-D scene flow	62	324	23	69	478
P02	Target pose	77	331	17	46	471
P03	Scene flow + target pose	76	295	51	52	474
Scene replacements: target-pose objective retained						
P04	Wan latent + pose	79	310	41	47	477
P05	Dense 2-D displacement + pose	87	356	14	57	514
P06	Codec motion + pose	96	250	23	96	465
P07	Sparse scene tracks + pose	47	329	20	57	453
Target replacements: scene-flow objective retained						
P08	Scene flow + object DINO	61	322	42	41	466
P09	Scene flow + target tracks	67	299	35	51	452
Additional objectives: scene flow + target pose retained						
A_α	P03 + motion magnitude	57	315	65	45	482
P12	P03 + depth	62	327	40	40	469
P13	P03 + DINO	71	326	27	48	472
P14	P03 + dynamic RGB	70	322	18	60	470
P15	P03 + depth / DINO / dynamic RGB	59	343	19	37	458

Table S15. Future–encoder combinations at seed 0: successes /1,400 (the development selection score, including fixed140). Rec. denotes current-target RGB reconstruction. All conditions receive a full visual window and GT target motion; DINO-S is frozen and the CNN trains from scratch. † marks selection by the development score. Both matched controls are complete. Dashes in the control rows denote unrun combinations. Appendix D gives task counts and paired outcomes.

	DINO-S		CNN	
Future supervision	No rec.	+rec.	No rec.	+rec.
P15: P03 + three targets	873	836	875	867
P07: scene tracks + pose	840	849	896	935†
P09: flow + target tracks	820	830	829	870
Matched controls for the selected configuration				
P00: no future	—	—	—	810
P03: flow + pose	—	—	—	861

D.3. Combination Selection

The combination screen crosses three future choices, two visual encoders, and current-target RGB reconstruction on/off. It preserves a full visual window and privileged target-motion input, disables Alpha and late injection, and trains each condition fresh for four epochs with the full-run cosine schedule. Its twelve candidates and the selected configuration's matched P00 and P03 controls are complete. All fourteen conditions start with sixteen devices and global batch 16. L and N resume on eight devices with batch 2 per device, preserving the optimizer, schedule, normalization, sample cursor, and 40,104-update budget, but resetting destination-rank random states. K resumes on a different GPU type with sixteen devices. M completes training on sixteen devices and moves only its remaining 1,260-scene evaluation to eight. I, E, and F retain sixteen-device training. The full grid thus matches nominal exposure, but its P09 candidates do not share one hardware and random-state history.

CNN and temporal conditioning. The trainable CNN has four convolution–GroupNorm–GELU blocks with channels $3 \to 32 \to 64 \to 128 \to 192$. Each convolution has kernel 4, stride 2, padding 1, and no bias; GroupNorm uses eight groups. The 224×224 image becomes 14×14 patches, followed by token LayerNorm ($\epsilon = 10^{-6}$) and a $192 \to 768$ projection. RGB is divided by 255, bilinearly resized with antialiasing, and normalized by means $(0.485, 0.456, 0.406)$ and standard deviations $(0.229, 0.224, 0.225)$. Geometry embeddings follow Appendix B. The full T8 window retains $8 \times 196 = 1,568$ visual tokens, eight EEF tokens, and eight object-kinematic tokens: 1,584 scene tokens when the full window is available, plus sixteen learned meta tokens. Relative-time embeddings and trunk attention fuse the window; there is no history-query compression or temporal mean pooling.

Current-target reconstruction. The mean of the final sixteen meta tokens enters a $768 \to 256 \to 3,072$ MLP with GELU and final sigmoid, producing $3 \times 32 \times 32$ RGB. Simulator target geometry IDs select the currently visible target and its tight bounding box. The crop is resized to 32×32 (bilinear RGB, nearest-neighbor mask); values outside the mask are zero. Loss is masked L1 with outer weight 0.1, divided by the global count of valid RGB elements. An empty mask contributes zero without dropping the training sample. The reconstruction is a training readout, not an action input. I, E, and F share this path, the same D backbone, and the same training normalization from 1,324,374 valid issued actions in 11,200 training episodes. The 350 validation episodes do not contribute normalization statistics.

Candidate I obtains 935/1,400 by the predefined selection score. Against J, the same P07/CNN condition without reconstruction, the paired counts are 817 joint successes, 118 I-only successes, 79 J-only successes, and 386 joint failures. Its net 39-scene gain is distributed unevenly over tasks, as Table S16 shows. The multi-seed combination matrix in Appendix E tests whether this development ranking persists.

The matched P00 control F retains I's trainable CNN, full visual window, privileged motion input, and current-target masked RGB reconstruction (weight 0.1), with future objectives disabled. Both use fresh seed 0, global batch 16, four epochs, and the full-run cosine schedule. F obtains 79/140 and 731/1,260 successes, totaling 810/1,400. For I versus F, the paired counts are 746 joint successes, 189 I-only successes, 64 F-only successes, and 401 joint failures. The 125-scene net gain is 8.93 percentage points (95% pointwise interval 7.00–10.86). It includes 72 on phase passage and 22 on replanning; translational placement loses seven. These are results on the pool used to select I; the interval does not correct for candidate selection.

The matched P03 control E changes the future objectives to scene flow and target pose. It obtains 87/140 and 774/1,260 successes, totaling 861/1,400. I versus E has 786 joint successes, 149 I-only successes, 75 E-only successes, and 390 joint failures. The net 74-scene gain is 5.29 percentage points (95% pointwise interval 3.50–7.07). Phase passage contributes 72 of these net successes; the remaining thirteen tasks contribute two in total. The aggregate difference is therefore highly task-concentrated within this development pool. A post-hoc calculation excluding phase passage gives 863/1,300 for I and 861/1,300 for E: 0.15 points (interval $[-1.69, 2.00]$). This descriptive sensitivity analysis does not replace the predefined fourteen-task score. Independent confirmation uses the new scene partition and seeds specified in Appendix A.

Table S16. Combination-study successes, grouped by future target and visual encoder. DINO is frozen DINO-S; CNN is trained from scratch. Minus/plus denotes current-target RGB reconstruction off/on; parentheses retain source condition IDs. E and F are the selected configuration's matched scene-flow/pose and no-future controls. Each task has one hundred scenes, including development scenes, and each condition has one training seed. Totals use the predefined 1,400-scene selection pool.

	P15: auxiliary				P07: scene tracks				P09: target tracks				P03	P00
Task	DINO − (B)	DINO + (A)	CNN − (D)	CNN + (C)	DINO − (H)	DINO + (G)	CNN − (J)	CNN + (I)	DINO − (L)	DINO + (K)	CNN − (N)	CNN + (M)	CNN + (E)	CNN + (F)
Reference grasp														
Grasp	90	89	91	92	90	92	87	90	79	85	77	87	85	78
T1: target motion														
Belt	100	100	99	100	100	100	100	97	99	100	90	99	100	100
Occlusion	100	100	99	99	99	100	100	97	96	99	82	98	99	98
SO(3)	86	71	80	88	75	87	69	86	69	62	64	78	78	71
XYZ	58	58	82	75	73	75	89	82	44	56	82	84	78	82
T2: receptacle motion														
SE(3)	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Translation	8	4	6	5	9	1	14	4	4	5	9	14	9	11
T3: constrained passage														
Full	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Phase	0	0	0	0	0	0	28	72	0	0	0	2	0	0
T4: object capture														
Can	99	99	99	98	100	99	99	100	97	99	96	99	99	99
Slider	88	85	88	91	74	83	86	81	84	84	88	88	84	77
Sphere	71	67	67	63	47	40	53	56	68	68	62	64	68	45
Tabletop sphere	100	100	100	99	99	100	95	98	100	100	100	100	99	99
T5: post-impact replanning														
Replan	73	63	64	57	74	72	76	72	80	72	79	57	62	50
Total / 1,400	873	836	875	867	840	849	896	935	820	830	829	870	861	810

D.4. Granularity and Prediction-to-Action Evidence

The role/allocation comparison fixes one 128-point scene-track representation, the current target identity, eight 50–400 ms horizons, and 128 future-token slots. A common current-frame candidate pool partitions valid source pixels by the same target mask into target and context. Each partition is ordered by deterministic farthest-point sampling in normalized image coordinates, starting at its centroid-nearest pixel and breaking ties in row-major order. Allocations take prefixes of these same ordered lists: 64/64, 96/32, or 32/96 target/context points. Insufficient points use masked padding; slots are not silently transferred to the other role. The current target mask and source roles are available in every row. No future motion magnitude selects a source point.

The 64/64 reference additionally has target-only and context-only supervision variants. They preserve both source sets and computation, changing only which role contributes coordinate/visibility losses. The fully supervised rows average the two role losses equally so that denser sampling does not also increase that role's objective weight; each loss averages its own valid points. Pose supervision, history, backbone, Action loss, and exposure remain fixed. All six rows use three training seeds. Thus role coverage and spatial allocation are distinct tests, and neither is inferred from the cross-representation P00–P15 screen.

Physical forecast errors use a common evaluation pool of 256 target and 256 context source points, obtained from the same deterministic ordering and excluded from the training queries. At evaluation the trained projection is queried at these points in four batches of 128 without action rollout; forecast quality and deployment latency therefore have distinct measurement passes. Translation/rotation readouts retain their native metric/geodesic units. Tracks report per-horizon coordinate error, visibility BCE, and valid coverage. Free-generation error is distinct from FM validation loss. The prediction-to-contact analysis aligns query, forecast, actual state, action application, and first contact on physical timestamps. Oracle and future-swap tests use the interface in Appendix E; task and safety outcomes accompany action changes.

Table S17. Role and allocation study within one 128-point future-track representation. Every row shares source candidates, current target identity, eight horizons, 128 slots, backbone, and exposure. T/C denotes target/context points; role-only rows mask only supervision. Errors use one common held-out source-point pool.

Condition	T/C	Supervision	SR (%)	Target px	Context px	p95 (ms)
Uniform	64/64	T+C	—	—	—	—
Shared bank	64/64	T+C	—	—	—	—
Target fine	96/32	T+C	—	—	—	—
Context fine	32/96	T+C	—	—	—	—
Target only	64/64	T	—	—	—	—
Context only	64/64	C	—	—	—	—

E. Backbone, Information Flow, and Design Confirmation

E.1. Parameter Organization

The backbone comparison holds history H, future representation Z, and reference information flow G_0 fixed. In G_0, future supervision updates shared parameters, but Action does not read a generated future. D shares attention QKV/output projections across observation/meta, future, and Action, with separate feed-forward networks. Adapted E has two parameter groups, World and Action, each with its own QKV, output projection, adaptive layer normalization (AdaLN), and feed-forward network. Observation/meta and active future tokens share World parameters. Observation/meta tokens use diffusion time $\tau = 0$; generated future tokens retain their own time. When the future is only a deterministic readout, neither backbone instantiates an unused generative stream.

Internal normalization is part of this comparison. E's World AdaLN has randomly initialized scale/shift biases at $\tau = 0$, whereas D's observation LayerNorm starts with affine scale one and shift zero. D/E therefore compares parameter organization and normalization as complete backbone choices. The early E system, with different inputs and a joint program, remains a separate system reference.

Both arms use fresh training seeds 0, 1, and 2. Common input projections, history/time parameters, and readouts start identically within a seed; backbone groups use separate initialization streams. The reference recipe is four epochs, global batch 16, 40,104 updates, and 641,664 consumed chunks per run (Appendix B). This gives six endpoints and 8,400 development rollouts. D uses FFN width $4d$, with $d = 768$. E chooses its FFN width from positive multiples of 64 to minimize the absolute active-backbone parameter difference, breaking ties toward the smaller width. The difference must be within 5%; depth, attention width, and heads remain fixed. Frontend, teacher, and total parameters are listed separately. Equal exposure does not imply equal FLOPs.

Selection rule. The fixed reference device is the previously measured NVML-identified `NVIDIA RPBZZZ6`, capacity 102,641,958,912 bytes, with FP32/TF32. All candidates use this same device and precision, with p95 time-to-action at most 200 ms and peak memory at most 80% of device capacity. For candidate minus reference SR, a 95% paired interval wholly above $+2$ percentage points supports a practically meaningful advantage; an interval inside $[-2, 2]$ supports the specified equivalence criterion. A candidate with at least 10% lower p95 latency can be selected on cost if its SR interval lies above -2 points. Per-seed scores and native task/safety outcomes accompany the decision. If three seeds do not resolve it, both arms add seeds 3 and 4. If neither dominates after this extension, the reference is retained without a superiority claim. The default complete reference is T8 RGB with causal estimated motion, scene tracks plus target-pose supervision, D, and R1 action-only deployment. Oracle-information studies retain their separate input track. Table S18 records the backbone comparison and its revisit under the selected information flow.

E.2. A Common Generated-Future Interface

Every information-flow row binds the same generated state: scene displacement $F \in \mathbb{R}^{196\times8\times3}$, or sparse image tracks $F \in \mathbb{R}^{128\times8\times2}$, over 50–400 ms. Displacements use the fixed robot-base frame. Tracks retain current source identities and use image coordinates divided by width and height. Source coordinates and current validity are observation inputs. Future values are standardized per coordinate channel over valid training elements, with a minimum scale of 10^{-6}; all rows use the same statistics. Each source point's full trajectory is projected to one token, with source-location and representation embeddings; the output projection reconstructs its eight ordered horizons.

For standardized trajectory x_1 and $\epsilon \sim \mathcal{N}(0, I)$, $x_\tau = (1-\tau)\epsilon + \tau x_1$ has FM target $x_1 - \epsilon$. One $\tau \sim U[0, 1)$ is shared within each future tensor. Coordinate MSE averages valid elements. Invalid future coordinates retain sampled source noise throughout interpolation and are excluded from loss; their masks do not condition the model. Current source

Table S18. Backbone comparison and directed revisit. SR averages common development scenes and training seeds. Parameter counts cover the active backbone; p95 is time to the first issued action. The revisit fixes the selected information flow.

Flow	Backbone	Params (M)	SR (%)	ΔSR (pp)	p95 (ms)	Memory (GiB)
G_0	D	—	—	ref.	—	—
G_0	E	—	—	—	—	—
$G^\star$	D	—	—	ref.	—	—
$G^\star$	E	—	—	—	—	—

padding is masked identically at training and deployment. A separate logit head predicts track visibility with BCE on valid correspondences. Ground-truth future visibility never enters the generator or Action. Target pose remains a deterministic meta-token readout with the geometric loss in Appendix B; it is not the generated state F. The World-to-Action edge exposes future tokens and their hidden states.

This normalized Gaussian interface is common to the information-flow comparison. The earlier representation screen used a 0.02 m metric scene-flow source and deterministic track readouts, so its checkpoints do not replace matched information-flow training. Auxiliary weights are fixed across R1–R6 and Alpha is zero.

E.3. Seven Information-Flow Programs

Let C denote observation/meta context, F generated future tokens, and A the 8×7 action state. Context is causal in capture time and cannot read F or A. Each generated stream reads C and itself; Table S19 gives the additional cross-stream edges. Future labels therefore cannot reach Action indirectly through context. A blocked edge has attention logit $-\infty$ at every layer and head in training and testing.

Synchronous programs. R0 removes future loss and tokens, preserving the action/context interface. R1 adds future supervision with both cross-stream edges blocked; its same checkpoint runs action-only or generates an unused future. R2 enables $F \to A$, R3 enables $A \to F$, and R4 enables both. Training uses independent Gaussian sources and independent uniform diffusion times for F and A, with per-stream time embeddings. At deployment both times advance on $\{0, .2, .4, .6, .8\}$: five combined evaluations. Streams exchange their noisy states and hidden representations, not clean future labels.

Sequential programs. R5 first generates F with Action absent, then holds its decoded tokens fixed while generating A through $F \to A$. Training uses two passes: future FM with Action absent, then action FM conditioned on clean ground-truth future tokens. The second pass does not backpropagate through a sampled first pass; both losses update shared parameters normally. R6 reverses these roles: action FM, then future FM conditioned on clean actions. At deployment the second stage reads first-stage predictions. Each stage uses five Euler steps, totaling ten evaluations. The training/deployment conditioning difference belongs to the sequential program. R6 can issue its first-stage action immediately; both time-to-action and time-to-complete-forecast are reported.

Three seeds give 21 trained endpoints and 24 deployment conditions, including R1's second mode: 33,600 development rollouts. Every row issues one command per replan with the same physical limits. World/Action NFE, forward calls, active parameters, exposure, p50/p95 time-to-action, memory, and physical forecast errors accompany SR. R0/R1 tests supervision; R1's two deployments isolate unused generation; R1–R4 tests access; R2/R5 and R3/R6 compare complete synchronous and sequential programs.

E.4. Interventions and Interaction Tests

Future interventions hold current perception, robot history, sampling arrays, and decoding fixed. Identity replacement copies native future states. Zero-motion replacement uses zero 3D displacement or repeats each current image point across horizons. A matched-current shuffle samples a different development trajectory with seed 17 in the same task, object, and visibility stratum, within 5 mm and 5° of current target pose. Donor sources also match receiver coordinates within two pixels and have the same role. Both members retain the same valid point/horizon domain; unmatched windows are excluded with coverage reported. Time shifts use $\delta_h \pm 50$ ms where labels exist. Noise uses standard deviations $0, 0.5, 1, 2$ times development prediction RMSE, with shared random draws and the representation's physical coordinates.

An oracle diagnostic uses measured exogenous target motion before first contact. Post-contact analysis forks a simulator snapshot under a fixed candidate action sequence and the same external motion. It measures response to that candidate's future; an action changed by the intervention need not produce that same future. R2/R5 model demonstration futures without action conditioning, whereas R3/R4/R6 can condition on action. Geometric validity alone does not certify action compatibility. Task and safety outcomes accompany first-issued translation and SO(3) action changes.

For R2/R4, a clean replacement is standardized using the

Table S19. Controlled information-flow programs. F denotes a generated future state and A an action state. Arrows specify information access, independently of whether the future loss is enabled. R1 has two deployments of the same trained checkpoint. Sequential second stages use clean ground truth in training and generated first-stage values at test time.

ID	Future loss	$F \to A$	$A \to F$	Cross-stream training input	Inference order
R0	No	—	—	None	A only
R1	Yes	—	—	None	A only
R1	Yes	—	—	None	Joint F and A
R2	Yes	Yes	—	Synchronous noisy states	Joint F and A
R3	Yes	—	Yes	Synchronous noisy states	Joint F and A
R4	Yes	Yes	Yes	Synchronous noisy states	Joint F and A
R5	Yes	Yes	—	Clean F for second-stage A	F, then A
R6	Yes	—	Yes	Clean A for second-stage F	A, then F

Table S20. Information-flow results. Each condition uses three training seeds and the same history/future interface. NFE counts are per modality; joint rows share calls. Forecast error is decoded to mm for 3D displacement or pixels for tracks, fixing representation within a comparison. An absent generated output is marked n/a.

ID	Deployment	NFE_F	NFE_A	SR (%)	Forecast error	p95 (ms)	Safety (%)
R0	Action	0	5	—	n/a	—	—
R1	Action	0	5	—	n/a	—	—
R1	Joint, unread future	5	5	—	—	—	—
R2	Joint $F \to A$	5	5	—	—	—	—
R3	Joint $A \to F$	5	5	—	—	—	—
R4	Joint, bidirectional	5	5	—	—	—	—
R5	F then A	5	5	—	—	—	—
R6	A then F	5	5	—	—	—	—

original training statistics and inserted at every Euler step as $F_u^I = (1-u)\epsilon_F + uF^I$, retaining the native source-noise array. World hidden states are recomputed before Action is evaluated; R4 keeps its current Action-to-World edge. Identity controls instead copy native intermediate F_u states and test exact reconstruction of the unmodified program. R5 replaces and re-embeds the clean decoded future at its second stage. R0/R1 and R3/R6, whose issued actions do not read future states, serve as negative controls. Label validity defines paired analysis support but is not an input mask. Table S21 reports action, contact, task, and coverage measures.

The confirmation matrix contains three interactions. First, current-only versus T8 RGB is crossed with current target pose versus pose plus interval motion, using the selected future target. Second, scene flow versus sparse tracks is crossed with G_0 versus the selected flow, retaining target-pose supervision. Third, D and E are revisited under that flow. Every cell uses seeds 0–2 and common development scenes; identical endpoints are reused. A ranking reversal is reported as an interaction and changes selection of the complete configuration. The selected (H, Z, B, G) tuple, its action-only ablation, and the strongest complete alternative are evaluated on the 1,400 independent confirmation scenes in Appendix A, without further selection.

Rotation interface. The proprioception/action ablation compares rotation vectors with fixed-axis XYZ Euler angles, using $R = R_z(\gamma)R_y(\beta)R_x(\alpha)$, $\beta \in [-\pi/2, \pi/2]$, and principal branches for the other angles. Proprioception is absolute orientation; Action is a left-relative increment, $R_{t+1} = \Delta R R_t$. Both encodings change together while physical frames, commands, data, and budgets remain fixed. SR and SO(3) geodesic error provide coordinate-independent evaluation. This is separate from the target-pose readout's encoding.

Table S21. Frozen future-interface interventions. Action changes compare first-issued commands under matched noise. Contact errors use actual contact events; non-contact failures stay in SR and are excluded only from conditional contact metrics. Coverage is the fraction of eligible windows/scenes with paired intervention support.

Intervention	Δa (mm)	ΔR (°)	Contact (ms)	Pose (mm)	SR (%)	Coverage (%)
Native future	0	0	—	—	—	—
Identity copy	—	—	—	—	—	—
Zero motion	—	—	—	—	—	—
Matched shuffle	—	—	—	—	—	—
Time offset ± 50 ms	—	—	—	—	—	—
Noise $0.5/1/2\times$ RMSE	—	—	—	—	—	—
Pre-contact oracle	—	—	—	—	—	—
Candidate-conditioned oracle	—	—	—	—	—	—

Table S22. Targeted interactions and independent confirmation. History rows fix the future design; future/flow rows retain target-pose supervision. Development contrasts match seeds and scenes. ID confirmation uses 1,400 new scenes after configuration selection.

Comparison	Condition	SR (%)	ΔSR (pp)	p95 (ms)
$H_{\text{RGB}} \times H_{\text{motion}}$	Current RGB / current pose	—	ref.	—
	T8 RGB / current pose	—	—	—
	Current RGB / interval motion	—	—	—
	T8 RGB / interval motion	—	—	—
$Z \times G$	Scene flow / G_0	—	ref.	—
	Scene flow / $G^\star$	—	—	—
	Sparse tracks / G_0	—	—	—
	Sparse tracks / $G^\star$	—	—	—
ID confirmation	Selected complete design	—	ref.	—
	Matched action-only ablation	—	—	—
	Strongest complete alternative	—	—	—

F. Robot-Data Pretraining, Transfer, and Deployment

F.1. Robot-Data Semantics and Partitions

The robot-data base uses DROID demonstrations [4], with the selected history, future targets, and information flow fixed. The training unit is a causal observation window and its aligned future/action labels. All windows from one episode remain in one partition. Collection session and scene groups are assigned together to training/development/held-out proportions 90/5/5 using seed 20261018; group-size rounding is reported by episode count. Groups that share a downstream test scene are excluded from pretraining. The split is formed before label filtering. Nested 12.5/25/50/100% training subsets preserve task/scene strata and use only their own normalization statistics. Counts of episodes, valid windows, unique actions, and valid labels are separate quantities (Table S23).

Time and action alignment. Source capture timestamps define history and future times. A nominal 50 ms grid is aligned field by field. Causal inputs use only arrived prefix samples: RGB and state retain their latest emitted value at or before the query, with actual capture age and validity. Previously emitted history is never reconstructed from a later sample. Only offline labels use linear position interpolation and shortest-path SLERP between valid bracketing samples; commands are held over their recorded application interval. No offline interpolation crosses a gap longer than two source periods, an object-identity change, or a reference-frame reset. Every resulting field retains its source times and validity. Repeating a frame or relabeling a nominal 15 Hz sequence as 20 Hz does not change its physical timestamp. Motion labels use these actual intervals.

Robot states and Cartesian commands are expressed in the robot-base frame, with the calibrated rigid flange-to-TCP transform applied before computing increments. The E7 command is the translation difference and left-relative rotation from measured current TCP pose to the commanded TCP pose, plus the source command's mapped gripper state. Measured gripper opening remains a state variable. Valid commands alone determine training normalization; over-

lapping windows do not count the same issued command repeatedly. Out-of-range commands retain a rejection reason and are not silently clipped into valid training labels.

Causal inputs and offline labels. DROID does not supply object-pose ground truth. The target is the task-specified manipulated rigid body, with one persistent instance and object coordinate frame per manipulation segment. A target prompt becomes available only at its recorded time. Causal estimation uses allowed views, past/current frames, and geometry available by that time. Forward tracking retains the original emitted poses and availability times. Lost tracking sets input validity false; re-localization starts a new segment unless identity and frame alignment are recovered.

The fixed frontend uses SAM 2.1 Hiera-small for causal mask propagation from a first-frame target box, followed by FoundationPose RGB-D registration and tracking with its released refiner/scorer weights [10, 14]. Both networks are frozen. The CAD model or independent reference views are available before the episode; geometry reconstructed from that episode's future is not an input prior. Each query processes only the newly available prefix, emits the current mask/pose, and appends the estimate to history without rewriting earlier emissions. Current-depth validity, tracker loss, and reference-frame changes propagate to object-state validity. A missing state uses a learned missing-object embedding plus a zero validity bit instead of normalized pose values. Training uses the same causal estimates and naturally occurring missing states, with no synthetic input dropout. The explicit missing-input path allows action/scene supervision when an object prior or pose is unavailable; its pose objective is masked. The full-pose subset is reported separately. Initialization prompting/registration and per-frame tracking costs enter the end-to-end runtime account. Pure-RGB external adaptations have a separate input condition and do not receive substituted oracle depth/pose.

Offline supervision uses the complete synchronized segment, DROID's released camera calibration, the recorded target prompt, and the same independently supplied object prior. SAM 2 propagates the mask forward and backward; a frame is accepted when mask IoU is at least 0.5. FoundationPose registers accepted RGB-D frames to the common object frame and refines forward and backward tracks. Frames with fewer than sixteen valid depth pixels, nonfinite transforms, or forward/backward disagreement above 20 mm or 15° are invalid. No object prior means no full-pose label.

The fused trajectory minimizes a Huber objective over the two pose estimates and adjacent interval-velocity changes. Pose residuals are translation divided by 5 mm and SO(3) angle divided by 3°; the Huber transition is one. Smoothness residuals are $\|v_t - v_{t-1}\|\Delta t/(5\,\mathrm{mm})$ and $\|\omega_t - \omega_{t-1}\|\Delta t/(3^\circ)$, with weight 0.1. They are omitted across gaps or initial-trajectory velocity jumps above 0.2 m/s or 1 rad/s to preserve abrupt events. Initialization uses the translation mean and SO(3) geodesic midpoint of the two estimates. Twenty manifold Gauss–Newton iterations keep this validity/edge set fixed. The teacher never rewrites emitted inputs. Future targets are $\tilde{p}_{t+h} - \tilde{p}_t$ and $\mathrm{Log}(\tilde{R}_{t+h}\tilde{R}_t^\top)^\vee$ from the offline trajectory, rather than subtracting a causal current estimate from an offline future. This avoids mixing current-pose correction into the motion target. Per-horizon validity requires a valid current reference, valid future pose, common identity/frame, and observable rotation. Missing supervision masks its loss; unavailable causal pose uses the explicit missing-input path. The complete SE(3) pool uses observable rigid-body poses under the ordinary SO(3) loss. Partial rotation or symmetry-class labels are distinct data conditions, not silently valid full-pose labels.

Label quality. The quality study uses 40 clips, 14 for development and 26 held out, with five anchors per clip. Seven strata cover stationary motion, smooth translation, rotation, contact/slip, occlusion/recovery, weak appearance, and symmetry or rigid articulated parts; counts per stratum are reported. The held-out clips do not tune the estimator. Prefix truncation and future replacement verify that emitted inputs, validity, target identity, and reference geometry are unchanged within repeated-run numerical tolerance. Independent manual landmarks and calibrated multiview geometry supply references where sufficiently constrained; otherwise physical pose accuracy is marked unverified.

Causal and offline paths report translation/rotation median and p95, horizon-wise motion error, identity failures, false accepts, retained coverage, recovery time, and compute/annotation cost. The contact-error budget is expressed as $e_p + 2r\sin(e_R/2) + ve_t$, where r is the contact lever arm and v the relative approach speed. Development tolerances are set from the task's geometric clearance and measured timing budget before held-out evaluation. This is an engineering bound, not a joint probability guarantee. Supervision quality and usable coverage are both reported.

Scene-track labels. Frozen DPFlow estimates bidirectional adjacent-frame flow; bilinear composition follows fixed current source points to the eight physical target times [8]. Forward/backward agreement uses the pixel-space criterion in Appendix C. A chain is valid only while its sampled coordinates and reverse checks are valid. MoGe-2 supplies metric pseudo-depth [12]; resized intrinsics and released extrinsics backproject correspondences for 3D displacement. Pseudo-depth is not measured truth. Visibility and correspondence validity remain separate; invalid depth disables 3D supervision without manufacturing zero motion. All comparisons use the same source points, frame-time mapping, and teacher versions.

Table S23. Robot-data accounting. Splits are grouped before filtering. Coverage uses all candidate windows as its denominator; future validity is counted separately at each horizon. Causal-input accuracy and offline-label accuracy use independent references where available.

Partition	Episodes	Windows	Valid actions	Pose coverage (%)	Track coverage (%)
Train (90% groups)	—	—	—	—	—
Development (5%)	—	—	—	—	—
Held out (5%)	—	—	—	—	—
Held-out pose quality		Translation med./p95 (mm)		Rotation med./p95 (°)	
Causal input		—		—	
Offline teacher		—		—	

F.2. Training Recipe and Scale

Let $\eta_0 = 2 \times 10^{-4}$, $B_0 = 16$, and $T_0 = 641,664$ consumed windows denote the reference benchmark recipe. Its optimizer, clipping, and full-budget cosine schedule are given in Appendix B. A common 24-candidate recipe search uses learning-rate multipliers $\{0.5, 1, 2\}$, global-batch multipliers $\{1, 4\}$, exposure multipliers $\{1, 2\}$, and existing auxiliary-loss multipliers $\{0.5, 1\}$. Warm-up consumes the reference 16,000 samples. Each candidate has its own complete cosine schedule; extending a low-learning-rate tail is not the short-budget control. All 24 candidates receive 25% of their assigned exposure. Development SR selects six to complete their full schedules; the best two have three training seeds. The same search allocation applies to initialization controls. Appendix H gives the measured recipe study that motivates reporting seed and task variation.

For robot pretraining, one full-data pass consumes every valid training window once. The reference exposure T_{robot} is four such passes, with global batch 256 and the same AdamW settings. Warm-up occupies the first 2.5% of exposure, followed by cosine decay to 0.1 of the peak. The same 24-candidate search varies learning rate, batch, exposure, and existing loss weights around this recipe on the reference-sized model; the chosen settings define the full-data reference budget. These quantities refer to valid consumed windows, not raw video-frame count.

Model scale varies within the chosen architecture at approximately $0.5, 1, 2, 4$ times reference active-backbone parameters. At fixed width 768 and eight heads, depths 2, 4, 8, and 16 provide these four candidates; exact counts include projections and readouts. Representation, image resolution, and tokens remain fixed. All four use the full data pool. The $1\times$ and $2\times$ candidates additionally use the three smaller nested subsets, giving ten principal training curves. Smaller subsets repeat more often at fixed exposure. Each scale receives six local trials: learning-rate multipliers $\{0.5, 1, 2\}$ crossed with world-loss multipliers $\{0.5, 1\}$, with batch and exposure fixed at the selected reference settings.

Compute comparisons use $0.5, 1, 2$ times the reference full-data compute budget for the $1\times$ and $2\times$ models, matching measured training FLOPs where available and otherwise device-specific GPU-hours. Matched-budget schedules are trained separately when intermediate endpoints do not share that schedule. Transfer is evaluated at 25/50/75/100% of each full training exposure. Source-domain loss describes learning; downstream development SR and deployment constraints select the scale and one common base checkpoint. Each candidate receives one reference downstream exposure in each domain, with the same data and post-training seeds 0–2. Eligible candidates are ranked by the equal mean of simulation and real-domain task-macro development SR. Real development uses five disjoint object/initial-condition trials per task/speed/delay cell and seed; none is a final test trial. Three independent pretraining seeds are used for the reference scale and its principal action-only ablation. Larger one-seed curves are identified as such, and three post-training seeds from one base quantify post-training variation only.

F.3. Independent Post-Training and Ablations

Every simulation and real-robot branch initializes directly from the same selected robot-data checkpoint. Each creates its own optimizer, schedule, training-only normalization, and task-specific readout parameters. Simulation-adapted weights never initialize the real-robot branch. Incompatible action/state projections are reinitialized with the same seed for both controls; all other compatible parameters are loaded.

Robot-pretrained and no-robot-pretraining models share general visual initialization and downstream data. Each is evaluated both with the common downstream recipe and after the same 24-candidate search allocation. Nested demonstration fractions are 10/25/50/100%, preserving task coverage. Simulation uses the benchmark recipe as the reference; real-robot exposure is four passes over its accepted demonstrations. Each branch varies its own recipe under the same relative search grid and uses training seeds 0–2. Thus equal downstream budgets do not imply equal total compute; pretraining cost is listed separately.

The action-only pretraining control uses R0, transferring common parameters and giving newly introduced future parameters the same initialization. A distinct loss ablation retains the full information paths but removes explicit world

Table S24. Ten principal model/data curves. Depth changes at fixed width, heads, and tokenization. Data fractions are nested independent-episode subsets; consumed training exposure is common. Each transfer score uses independently initialized downstream branches from that row's base.

Scale	Depth	Data (%)	Params (M)	Sim SR (%)	Real SR (%)	GPU h
0.5×	2	100	—	—	—	—
1×	4	12.5	—	—	—	—
1×	4	25	—	—	—	—
1×	4	50	—	—	—	—
1×	4	100	—	—	—	—
2×	8	12.5	—	—	—	—
2×	8	25	—	—	—	—
2×	8	50	—	—	—	—
2×	8	100	—	—	—	—
4×	16	100	—	—	—	—

losses. When Action reads future states, action gradients or sequential teacher forcing can still use that path, so this latter condition is not a no-future-information control. The remaining comparisons are world supervision introduced only during post-training versus both stages; the selected pseudo-label filter versus an unfiltered quality control subsampled to the same number of valid labels; and frozen versus trainable compatible visual parameters. The two most consequential history/future interventions are repeated after pretraining. A data-diversity comparison applies only if an additional real-robot source is included, matching independent data amount and exposure. Table S25 keeps simulation, external, and real outcomes separate.

F.4. External Benchmarks and Real-Robot Tasks

Single-arm DOMINO [2] uses its native dynamic interception and tracking tasks, official training/development split, and success evaluator. All included tasks have 50 held-out scenes per training seed, with the native task count stated with the result. Native task success is averaged equally; task definitions and thresholds are not replaced by the internal fourteen-task evaluator. LIBERO [5] uses the Spatial, Object, Goal, and Long suites: ten tasks each, 50 evaluation scenes per task, and three training seeds. Demonstration and initial-state splits follow the released suite protocol. Reported scores remain separate by suite.

The reference platform contract is a seven-DoF torque-controlled arm, parallel gripper, calibrated fixed RGB-D camera, and timestamped Cartesian TCP interface. These are platform requirements; model names and measured calibration values are deployment metadata. Geometry uses independently measured object extent and receptacle/opening clearance. Trials last at most 15 s after target motion begins. Grasping requires lifting at least 50 mm above the initial support and retaining the object for 1 s. Placement requires full release with all object geometry inside the receptacle for 1 s. Passage requires all object geometry to cross the opening plane without obstacle contact, followed by 1 s retention. Re-interception requires an external impact followed by the same lift-and-hold grasp; robot contact before the impact fails. Apparatus dimensions follow the same object/clearance ratios for every method and are fixed before trials.

The deployment study uses four task families: moving-target grasping, placement into a moving receptacle, passage through a timed opening, and re-interception after an external impact. Each uses three speed levels $(0.5, 1, 1.5)v_{\mathrm{ref}}$ and three added command delays $(0, 1, 2)\Delta_c$. Here v_{ref} is two thirds of the maximum speed passing ten policy-independent motion-only calibration cycles without apparatus collision, workspace exit, or fixture instability. The declared device speed limit bounds a ten-step bisection search for that maximum. Δ_c is the measured controller period. Delay is added after inference and reported together with total action age. Ten object/initial-condition trials per cell give $4 \times 3 \times 3 \times 10 = 360$ trials per training seed, or 1,080 per method over seeds 0–2. All methods repeat the same trial conditions. The arm, gripper, camera, controller period, calibration, object dimensions, and absolute speeds form the deployment metadata, rather than assumed hardware facts.

Objects and background layouts used for final transfer testing are disjoint from downstream training and development. A fixed cyclic method order is rotated between successive trials to distribute drift; failures are reset before the next method. Human contact, safety stop, or manual task completion counts as failure. An infrastructure loss before a usable observation is repeated for every method in that paired block and counted separately. Success requires the physical task condition to persist for its prescribed hold duration. Collision, drop, workspace exit, intervention, and timeout are recorded separately. Contact-time error is relative to the feasible task contact window, and relative-pose

Table S25. Transfer and pretraining controls at the selected scale. All rows share downstream demonstrations and matched search allocation. Simulation, native external suites, and physical tasks retain distinct score denominators. The four LIBERO entries are Spatial/Object/Goal/Long.

Initialization / objective	Internal SR	DOMINO	LIBERO	Real SR
No robot pretraining	—	—	—	—
Action-only robot pretraining (R0)	—	—	—	—
Complete robot pretraining	—	—	—	—
Without explicit world loss; paths retained	—	—	—	—
Matched label-count quality control	—	—	—	—
Frozen visual encoder	—	—	—	—
Downstream data 10% / 25% / 50% / 100%	—	—	—	—

error is evaluated at first contact; non-contact failures remain in SR and are reported as missing contact measurements rather than zero error.

F.5. Delay and Cost Evaluation

The physical-delay sweep adds 0, 50, 100, 150, and 200 ms to the native action queue, preserving the task's existing delay. The main comparison uses 150 ms added delay. All methods use the same observation cutoff and queue; the task simulator continues evolving while the prescribed delay elapses. The sweep is distinct from wall-clock deployment, which includes capture, preprocessing, inference, queuing, and actuator application.

For each device/precision pair, timing uses thirty warm-up calls and 300 measured calls over the same three eight-frame observation sequences, with CUDA synchronization at call boundaries. Three timing rounds rotate method order. Time-to-action includes internal perception and action generation; complete-forecast time is recorded separately. External image I/O and simulator rendering are excluded from this microbenchmark. Deployment traces record sensor capture, availability, preprocessing, inference start/end, queue entry, command issue, and application. Action age is application minus source capture time. p50/p95/max age, deadline-miss fraction, peak allocated memory, and total resident memory are reported with device, precision, batch size, and sampling calls. The selection limits are p95 time-to-action ≤ 200 ms and memory $\leq 80\%$ of the fixed device capacity.

The original future screen's measured call means are 177.071–184.212 ms, p95 178.726–185.133 ms, and peak allocation 1.571–1.617 GiB. That protocol used six warm-up calls and thirty measurements over three sequences; seven real calls primed history before each timed eighth call. It used FP32/TF32, $B = S = 1$, five Euler action steps, CPU affinity 0–2, and a shared device reported by NVML as `NVIDIA RPBZZZ6` with 102,641,958,912 bytes. Loading, external oracle reads, rendering, I/O, and priming were excluded. These descriptive measurements retain their original sampling protocol.

F.6. Initialization Controls

Three auxiliary controls separate policy initialization from downstream design. DreamVLA removes its LIBERO policy checkpoint while retaining MAE/CLIP initialization, teacher targets, losses, and the independent twenty-epoch recipe (48,220 updates, batch 512). The $\pi_{0.5}$ control replaces base initialization with its LIBERO expert, keeping 30,000 updates and batch 256. Both retain the original fourteen-task partitions.

A third control uses action-only D (four layers, width 768), a trainable CNN, and T8 RGB/robot history, removing target pose, depth/camera geometry, future flow, and all target-region/future objectives. Fresh and LIBERO-90 initialized versions share normalization fitted on the same 11,200 demonstrations. LIBERO pretraining uses 4,050 training and 450 held-out demonstrations, thirty epochs, 8,520 updates, and batch 256. Both downstream runs reset optimizer/scheduler and use four epochs, 40,104 updates, and batch 16. This is a visual-motor transfer control for the simplified policy, separate from the formal robot-data base and language-conditioned LIBERO evaluation.

G. Diagnostics and Qualitative Evidence

G.1. Supervision at the Executed Action Position

All 800 historical full-passage demonstrations release only at the final action: chunk position h_7 has release supervision, while the executed position h_0 has none. This identifies an action-position coverage gap.

G.2. Gripper Execution Diagnostics

The gripper investigation compares sign decoding, a fixed 0.1 dead zone, and nearest-label ternary projection and history-hold mappings. The ternary projection maps to $\{-1, 0, +1\}$ with thresholds at ± 0.5. For the reported fifth-epoch endpoint, the 1,260-scene counts are 698 for sign, 742 for the fixed dead zone, 749 for nearest-label ternary projection,

Table S26. Delay sensitivity and physical deployment cost. The controlled delay is added to the native command queue; synchronous microbenchmark latency is not substituted for action age. SE(3)-WAM and its matched R0 control use the same estimator and device.

	SR (%) at added delay (ms)					Age p95	Miss (%)
Method	0	50	100	150	200	(ms)	
Matched R0	—	—	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—	—	—
Strongest system reference	—	—	—	—	—	—	—

Table S27. Auxiliary initialization controls on the original fourteen-task development partitions. Training exposure is fixed within each pair; these are separate from the robot-data-base study.

Policy	Initialization	140-scene SR (%)	1,260-scene SR (%)
DreamVLA	LIBERO policy; MAE/CLIP	—	—
DreamVLA	MAE/CLIP only	—	—
$\pi_{0.5}$	Base	—	—
$\pi_{0.5}$	LIBERO expert	—	—
Action-only D	Fresh	—	—
Action-only D	LIBERO-90 visual-motor	—	—

and 741 for history hold. The ternary projection does not show a clear overall improvement over the fixed dead zone; history hold adds failures outside the capture subset. The comparisons retain sign decoding as their reference.

Decoder and representation effects use separate comparisons. Appendix H gives the frozen-readout study, including its two adverse development cases.

G.3. Offline Target-Pose Predictions

Figure S5 compares deterministic fourth-epoch P15 pose readouts with two kinematic baselines on 84 states (six per task). All eight horizons are valid, yielding 672 state–horizon pairs. Quantiles use linear interpolation with equal state weights; states from one episode may be dependent. Median translation error rises from 1.45 mm at 50 ms to 4.04 mm at 400 ms; the corresponding rotation medians are $0.47°$ and $1.07°$. The cases are selected by mean translation error, independently of task success.

Kinematic baselines. CV and CA use the final two and three observed GT poses, respectively, within the same history available to P15. They do not read future labels or instantaneous simulator velocities. The saved raw positions, xyzw quaternions, and FP64 timestamps verify a 50 ms interval d for all states. The resulting last-interval derivatives agree with the model's unstandardized object-input values. For positions p_{-2}, p_{-1}, p_0 and a forecast horizon δ in seconds, CV predicts $\delta(p_0 - p_{-1})/d$. CA predicts $\delta v_0 + \delta^2 a/2$, where

$$v_0 = \frac{3p_0 - 4p_{-1} + p_{-2}}{2d}, \qquad a = \frac{p_0 - 2p_{-1} + p_{-2}}{d^2}.$$

CV retains the final interval velocity; CA extrapolates it from the -25 ms interval center to the current time.

In the fixed robot-base frame, define $\omega_+ = \mathrm{Log}(R_0 R_{-1}^\top)^\vee/d$ and $\omega_- = \mathrm{Log}(R_{-1}R_{-2}^\top)^\vee/d$. Rotational CV uses $\Delta R(\delta) = \mathrm{Exp}(\delta\omega_+^\wedge)$. CA sets $\omega_0 = 1.5\omega_+ - 0.5\omega_-$ and $\alpha = (\omega_+ - \omega_-)/d$, and integrates to $t = \delta$:

$$\dot{\Delta R}(t) = [\omega_0 + t\alpha]_\times \Delta R(t), \qquad \Delta R(0) = I.$$

Exponential midpoint steps are at most 1 ms; halving the step changes any predicted rotation by at most $0.000548°$. There is no smoothing, clipping, or fitted baseline parameter.

Table S28. DEV84 offline prediction errors: 84 states and eight horizons, yielding 672 dependent observations. CV/CA use the last two/three GT poses; P15 also uses the full history and its other observation inputs.

	Translation (mm)		Rotation (°)	
Method	Mean	Median	Mean	Median
P15	3.762	2.771	1.772	0.857
CV	12.943	1.444	9.675	0.175
CA	22.413	0.634	10.114	0.170

Table S28 reports both mean and median errors. P15 has lower means but higher pooled medians than either extrapolator: the baselines combine many small errors with a large-error tail. Using each state's mean over eight horizons, P15 has lower translation error than CV and CA on 42/84 and 37/84 states, and lower rotation error on 27/84 and 34/84. Thus the aggregate means do not establish uniform

prediction superiority. The comparison characterizes offline error distributions under GT motion inputs.

G.4. Closed-Loop Alignment

The qualitative protocol aligns observation capture, generated prediction, actual future, command application, and first contact on the physical clock. It includes the adverse P05 sphere condition, sparse trajectories, and the matched future interventions of Appendix E. For each condition, examples at the 50th and 90th percentile of per-episode prediction error are selected without task-outcome filtering; the largest action-change failure is included separately. Scene/camera identity, units, visibility, and terminal outcomes accompany every sequence. Simulation overlays use the stored camera calibration. Physical videos display measured capture and issue times and distinguish estimated object pose from independently measured task events. These selection rules retain both representative and adverse outcomes.

H. Additional Training and Execution Studies

H.1. Paired Training-Recipe Study

The recipe study tests two development combinations, G (frozen DINOv3-S) and I (trainable CNN), both with P07 supervision, the full visual window, and current-target RGB reconstruction. For each profile, training seeds 0 and 1 compare global batch 16 on sixteen GPUs against batch 64 on thirty-two GPUs. The larger-batch recipe doubles the learning rate from 2×10^{-4} to 4×10^{-4} and reduces warm-up from 1,000 to 250 updates, preserving 16,000 warm-up samples. This is a joint recipe and resource comparison, not an isolated batch-size intervention.

All eight runs complete four epochs and 641,664 samples. B16 uses 40,104 updates; B64 uses 10,028, retaining each epoch's 32-sample tail without padding, repetition, or dropping. AdamW, clipping, FP32/TF32, losses, normalization, and the four-epoch cosine exposure are held fixed. All final checkpoints undergo the same sixteen-device full350 validation and the original 1,400-scene evaluation with paired scene and inference seeds. The seed-0 B16 runs are fresh reruns: their scene-level success outcomes exactly reproduce the corresponding BASE2 endpoints. They do not add a new independent training seed. Seed 1 supplies the additional realization.

Performance and variability. For G, B64 minus B16 gives $+0.43$ and -0.86 percentage points at seeds 0 and 1, with paired 95% intervals $[-1.21, 2.07]$ and $[-2.50, 0.79]$. For I, the changes reverse direction: -3.64 $[-5.64, -1.64]$ and $+5.57$ $[3.57, 7.57]$ points. At B16, I falls from 935 to 857 successes between seeds; phase passage falls from 72 to 14. Neither the original seed-0 score nor the mean response to a recipe change establishes stability. The matched E/F controls have not been repeated at seed 1.

The two-seed mean recipe differences are -0.21 $[-1.36, 0.96]$ points for G and $+0.96$ $[-0.50, 2.39]$ for I. Intervals use 20,000 percentile bootstrap samples, RNG seed 20261017, resampling 100 scene identities within each task and retaining their batch and training-seed pairing. They condition on these two trained seeds and fourteen tasks; they do not estimate variation over new training seeds. Table S30 also shows task losses hidden by aggregate scores: G loses 17 XYZ successes at seed 0 and 15 SO(3) successes at seed 1. Auxiliary metrics are not uniformly preserved; for example, I's pose rotation error increases by 6.17% and 14.35%.

Time and adoption. Mean job-creation-to-result time decreases from 302.43 to 120.77 minutes for G and from 257.81 to 101.38 minutes for I. These measured times include queueing, setup, training, saving, validation, export, and evaluation. They do not identify a hardware-only speedup. Training-compute GPU-hours decrease from 70.22 to 52.47 for G and from 59.42 to 42.31 for I; this measure excludes saving and validation and is not a billing record. The predeclared working tolerance was a three-point aggregate SR loss. G satisfies it for both observed seeds, whereas I does not. The project therefore retains B16 as the general recipe; B64 is a conditional option for this G configuration when aggregate SR is the criterion. It is not adopted for I, task-sensitive comparisons, or an untested final design.

H.2. Frozen Three-State Gripper Readout

The execution study freezes the fifth-epoch P03 policy and learns a three-state readout. Its 889-dimensional input concatenates the current action hidden state at the final Euler evaluation ($\tau = 0.8$), eight frames of available end-effector pose, gripper opening and validity, the sampled six-dimensional arm-action horizon, and the current raw gripper value. A 128-hidden-unit GELU MLP predicts open/hold/close logits; argmax replaces only the executed slot-0 gripper command. A second head with the same hidden width reads only the raw gripper value. No contact truth or future label is added to the observation. The frozen policy, normalization, five Euler steps, and synchronous single-step execution remain common.

Supervision and offline evaluation. The original 11,200 training and 350 validation demonstrations supply 1,324,374 and 41,533 valid issued-command labels; training and validation source seeds are disjoint within each task. Every valid current command is retained, including windows whose later action slots are invalid. Only capture demonstrations contain hold labels: 2,562 in training (0.1934% of all labels) and 79 in validation. The other thirteen tasks have no valid hold labels; all phase-passage training and validation labels are

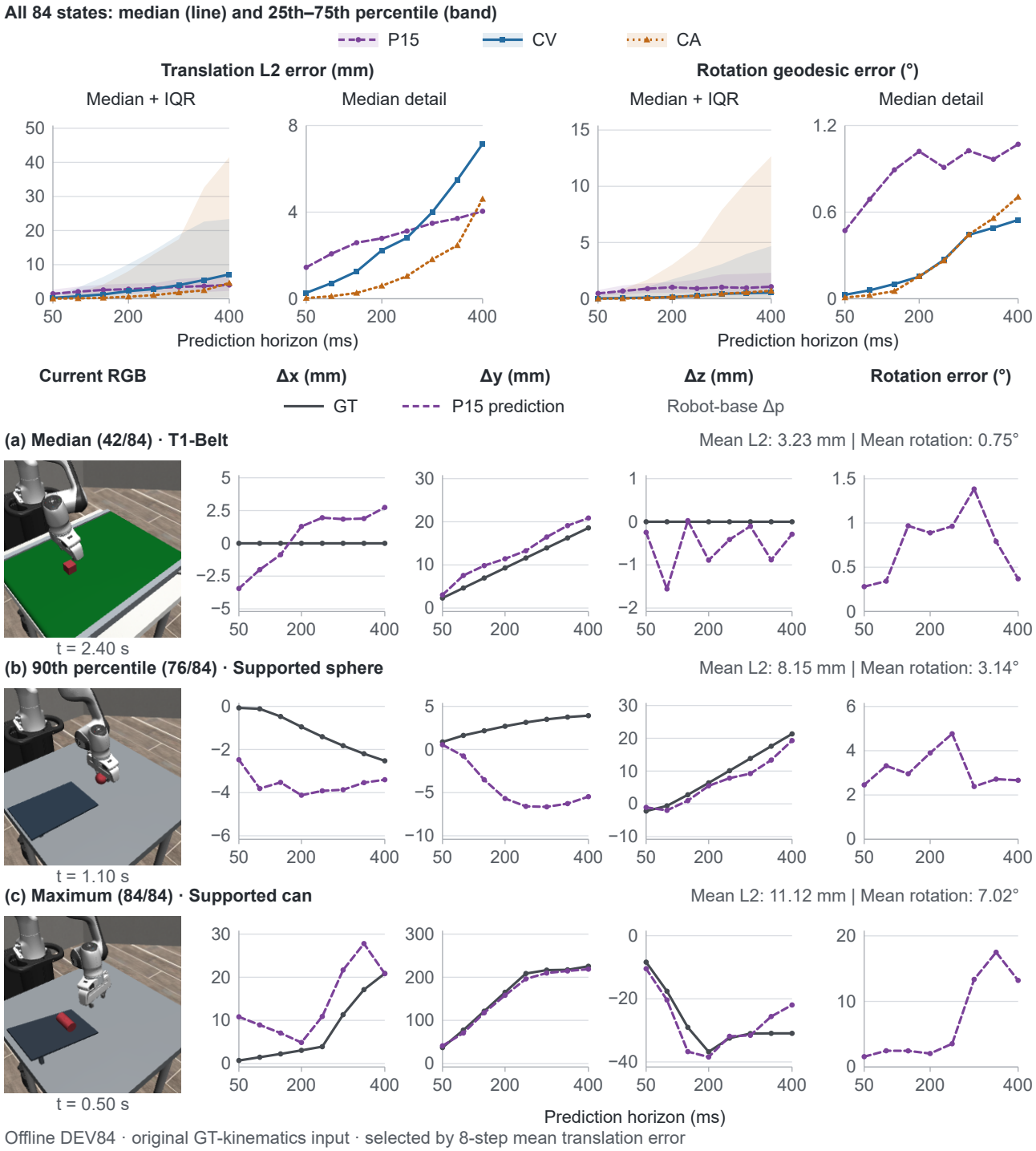


Figure S5. Offline P15 predictions and kinematic extrapolation. The DEV84 probe set contains 84 query states from 28 episodes across fourteen tasks. Top: P15, constant-velocity (CV), and constant-acceleration (CA) median translation L2 and rotation geodesic errors at each horizon; bands show the 25th–75th percentiles across states, not confidence intervals. The adjacent Median detail panels show the same median curves on expanded linear scales, without IQR bands. Bottom: we sort states by their mean translation L2 error over eight horizons and select nearest ranks 42, 76, and 84 (50th, 90th, and 100th percentiles), breaking ties by state index. Each row shows the original current image, predicted and true cumulative translation in robot-base coordinates, and rotation geodesic error at 50–400 ms. The model retains its GT target kinematics and robot-state history. These diagnostic examples characterize offline prediction errors; they do not measure closed-loop contact or task success.

Table S29. Completed training-recipe study, four epochs per run. G uses frozen DINOv3-S and I a trainable CNN; both use the corresponding P07 combination. Batch 64 jointly changes learning rate and device count (details in text). Training time excludes saving and validation. Job time runs from job creation through complete evaluation, including the observed queue and startup.

Profile	Seed	Batch	GPUs	Success /1,400	SR (%)	Train (min)	Job (min)
G	0	16	16	849	60.64	263.40	301.28
G	0	64	32	855	61.07	99.27	121.32
G	1	16	16	834	59.57	263.28	303.58
G	1	64	32	822	58.71	97.48	120.23
I	0	16	16	935	66.79	220.75	257.20
I	0	64	32	884	63.14	78.86	101.74
I	1	16	16	857	61.21	224.91	258.42
I	1	64	32	935	66.79	79.80	101.02

Table S30. Training-recipe successes per task (100 scenes per entry). B16 and B64 include their respective learning-rate/device settings; s0/s1 denote training seeds. All runs use the same accepted scene identities and inference seeds. Differences across tasks remain visible despite similar totals.

	G: frozen DINOv3-S				I: trainable CNN			
Task	B16/s0	B64/s0	B16/s1	B64/s1	B16/s0	B64/s0	B16/s1	B64/s1
Grasp	92	87	91	89	90	95	87	96
Belt	100	100	100	97	97	100	99	100
Occlusion	100	99	100	100	97	98	97	100
SO(3)	87	88	68	53	86	75	56	55
XYZ	75	58	60	69	82	69	77	85
SE(3) placement	0	0	0	0	0	0	1	0
Trans. placement	1	10	10	4	4	5	13	29
Full passage	0	0	0	0	0	0	0	0
Phase passage	0	0	13	0	72	32	14	46
Can	99	99	97	98	100	98	99	100
Slider	83	81	76	80	81	86	75	89
Sphere	40	50	50	57	56	47	60	73
Tabletop sphere	100	99	99	100	98	99	99	100
Replan	72	84	70	75	72	80	80	62
Total	849	855	834	822	935	884	857	935

close. This limits what the decoder can learn about useful holds outside capture.

Both heads use seed 0, twenty epochs, AdamW with learning rate 3×10^{-4} and weight decay 10^{-4}, batch 4,096, gradient clipping at 1, and ordinary cross-entropy without class reweighting. Feature means and variances use training data only. Each head is selected by the lowest validation mean of the three per-class NLLs, choosing the earliest epoch in a tie. Context selects epoch 15 with NLL 0.01330; raw-only selects epoch 20 with NLL 0.19640.

The context head recalls 79/79 hold labels with precision 79/83, versus 75/79 and 75/85 for raw-only (Table S31). It correctly classifies 137/153 one- or two-tick pulses, compared with 93/153; neither head incorrectly maps these pulses to hold. These counts describe frozen features on demonstration observations, not closed-loop policy performance. All three per-class NLLs improve. The 79 hold labels receive one third of the checkpoint-selection metric and account for 74.5% of its reduction; training cross-entropy

Table S31. Frozen-decoder validation confusion matrices: 41,533 valid issued commands from 350 demonstrations. Rows are ground truth; columns are predicted open/hold/close. These are offline classification counts, not policy successes.

	Raw only			Context		
GT	Open	Hold	Close	Open	Hold	Close
Open	8003	1	214	8132	0	86
Hold	0	75	4	0	79	0
Close	135	9	33092	105	4	33127

remains unweighted. The context head still makes 86 open-to-close and 105 close-to-open validation errors, so the mean loss does not constrain the cost of an individual closed-loop decision.

Predefined development cases. The context head is tested on eleven known development cases. It succeeds on E5 capture scenes 0/9/14/80/85, E5 replan scenes 49/88, and E7 XYZ scene 64 and replan scene 7; the E7 cases reuse the E5-trained head without further fitting. It fails on E5 capture scene 52 and replan scene 35. The head succeeds in 9/11 cases, below the predefined all-eleven criterion, and is therefore excluded from the final policy. The case count describes this selected development set.

An observation-only control preserves the original commands and random state. Twenty-eight paired sign/dead-zone trials reproduce their reference outcomes. Offline recomputation of 608 issued commands gives a maximum CPU-logit error of 3.82×10^{-6} and argmax commands agree exactly. The other six action dimensions and predicted horizons also pass the recorded comparisons. Complete per-frame policy-input tensors were not saved, so this is not a full input-tensor equality check.

Physical failure sequence. Figure S6 shows both failed cases. In each, the first intervention reverses an approximately -1 open command into $+1$ close; it is not merely suppression of a near-zero output. Capture changes at 0.650 s, followed by opening divergence at 0.666 s and no bilateral contact. Replan applies the altered command at 1.550 s, after its native one-tick queue; opening diverges at 1.556 s. The learned head fails at 1.324 s (tray contact) and 5.174 s (downstream exit), respectively. Sign succeeds in both; the fixed dead zone succeeds in replan but also fails in capture. Recorded predictions, commands, and 2 ms physical fields match their references before intervention. Subsequent arm and object divergence reflects closed-loop feedback. These records establish failures of this readout under the recorded conditions; they do not test a future-to-action prediction interface or establish decoder performance on unseen scenes.

Retained-feature diagnostics. At the first reversal in capture 52 and replan 35, replacing only the action hidden state with the preceding frame's value changes the close-minus-open logit margin from $+2.452/+0.700$ to $-13.839/-9.811$. Conversely, inserting the current hidden state into the preceding feature vector yields $+2.243/+3.184$; replacing only the raw value does not remove either reversal. These substitutions measure a local response of the frozen classifier to mixed features, which need not correspond to realizable observations or successful executions. The first reversal precedes any prior intervention by this head. The retained data do not separate demonstration-to-policy feature mismatch from sampling sensitivity of the final Euler hidden state; both explanations remain untested. The two failures remain unexplained by a single isolated mechanism.

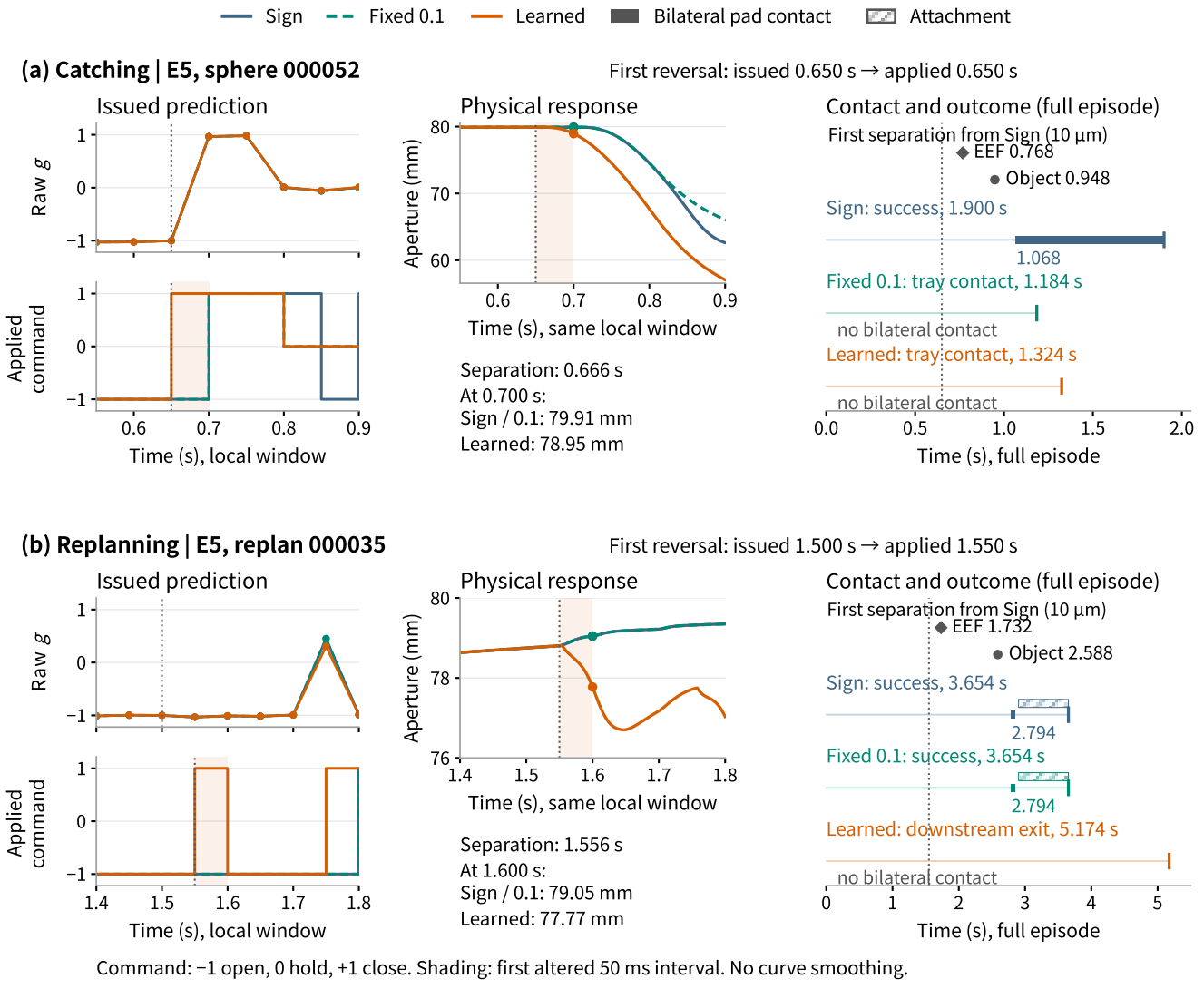


Figure S6. Frozen gripper-readout failures on the two failed predefined development cases. Commands and measured opening use physical time; event strips retain each run's contact and terminal outcome. Sign, the fixed 0.1 dead zone, and the learned context head use the same frozen policy. The head's first intervention reverses opening into closing, before the observed physical divergence. Detailed windows and full-run events serve different time ranges, labeled on their axes. Contact refers to actual bilateral collision; replan attachment is a separate event. Raw predictions are sampled at 20 Hz and physical states every 2 ms. Separation of the learned run from Sign uses a 10 μm aperture or componentwise position threshold. The cases are selected development diagnostics, not new-scene success estimates or future-prediction interventions.

References

[1] Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin C. M. Burchfiel, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Proceedings of Robotics: Science and Systems*, 2023. 8

[2] Heng Fang, Shangru Li, Shuhan Wang, Xuanyang Xi, Dingkang Liang, and Xiang Bai. Towards generalizable robotic manipulation in dynamic environments. In *European Conference on Computer Vision*, 2026. 25

[3] Gunnar Farnebäck. Two-frame motion estimation based on polynomial expansion. In *Proceedings of the Scandinavian Conference on Image Analysis*, pages 363–370, 2003. 9

[4] Alexander Khazatsky, Karl Pertsch, Suraj Nair, et al. DROID: A large-scale in-the-wild robot manipulation dataset. *arXiv preprint arXiv:2403.12945*, 2024. 22

[5] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. *arXiv preprint arXiv:2306.03310*, 2023. 25

[6] Yushan Liu, Jingjing Fan, Shoujie Li, Yifan Xie, Xiao-Ping Zhang, and Wenbo Ding. PACT-WAM: Predicting Actions and Visual Foresight with Compact Temporal Encoding for Robot Manipulation. *arXiv preprint arXiv:2602.15882*, 2026. 9

[7] Yunfan Lou, Hewen Gao, Xiyu Zhu, Zhuoran Qiao, Xuan Han, Yifan Yang, Yifan Ye, Boxian Yao, and Zhibo Pang. DynamicWAM: Dual-path motion conditioning for world-action models in dynamic manipulation. *arXiv preprint arXiv:2608.00793*, 2026. 7, 8

[8] Henrique Morimitsu, Xiaobin Zhu, Roberto M. Cesar, Xiangyang Ji, and Xu-Cheng Yin. DPFlow: Adaptive optical flow estimation with a dual-pyramid framework. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17810–17820, 2025. 23

[9] Physical Intelligence, Kevin Black, Noah Brown, James Darpinian, et al. $\pi_{0.5}$: a vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025. 6, 8

[10] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. SAM 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024. 23

[11] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, et al. DINOv3. *arXiv preprint arXiv:2508.10104*, 2025. 5

[12] Ruicheng Wang, Sicheng Xu, Yue Dong, Yu Deng, Jianfeng Xiang, Zelong Lv, Guangzhong Sun, Xin Tong, and Jiaolong Yang. MoGe-2: Accurate monocular geometry with metric scale and sharp details. *arXiv preprint arXiv:2507.02546*, 2025. 23

[13] Yuran Wang, Siqiao Huang, Mingleyang Li, Chenhao Zhang, Jiaqi Liang, Weiyang Jin, Yue Chen, Xuemin Chi, Donghao Zhou, Qize Yu, Yu-Kai Wang, Yuhan Rui, Shenzhe Yao, Zhen Yuan, Zhenhao Shen, Kefei Zhu, Zijie Zhu, Ning Gao, Xiaowei Chi, Guanqi He, Shanghang Zhang, Hao Dong, Lin Shao, and Hang Zhao. OpenWAM: An Open, Modular Exploration Towards Systematic World-Action Model Pretraining. *arXiv preprint arXiv:2609.07398*, 2026. 7, 8

[14] Bowen Wen, Wei Yang, Jan Kautz, and Stan Birchfield. FoundationPose: Unified 6d pose estimation and tracking of novel objects. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17868–17879, 2024. 23

[15] Haozhe Xie, Beichen Wen, Jiarui Zheng, Zhaoxi Chen, Fangzhou Hong, Haiwen Diao, and Ziwei Liu. DynamicVLA: A vision-language-action model for dynamic object manipulation. In *Advances in Neural Information Processing Systems*, 2026. 8

[16] Wenyao Zhang, Hongsi Liu, Zekun Qi, Yunnan Wang, Xin-Qiang Yu, Jiazhao Zhang, Runpei Dong, Jiawei He, He Wang, Zhizheng Zhang, Li Yi, Wenjun Zeng, and Xin Jin. DreamVLA: A vision-language-action model dreamed with comprehensive world knowledge. In *Advances in Neural Information Processing Systems*, pages 24195–24228, 2025. 7, 8

[17] Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning fine-grained bimanual manipulation with low-cost hardware. *arXiv preprint arXiv:2304.13705*, 2023. 8